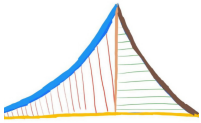
Matrices In Geometry



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018
First edition published on July 10, 2024

iIn this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. AE 2026

A. Formulae

1.1.1

$$\Pr(M) + \Pr(E) - \Pr(ME) + \Pr(M+E)' = 1 \tag{1.1.1.1}$$

1.1.2

$$A + A' = 1, \quad AA' = 0 \tag{1.1.2.1}$$

$$\implies \Pr(A + A') = \Pr(1) = 1 \quad \Pr(AA') = \Pr(0) = 0 \tag{1.1.2.2}$$

1.1.3

$$\Pr(A + A') = \Pr(A) + \Pr(A') \tag{1.1.3.1}$$

1.1.4 Difference rule of probability

$$\Pr(AB') = \Pr(A) - \Pr(AB) \tag{1.1.4.1}$$

Solution: From (1.1.2.1)

$$A = A \cdot 1 = A(B + B') = AB + AB' \tag{1.1.4.2}$$

Also,

$$(AB)(AB') = 0 \tag{1.1.4.3}$$

From (1.1.3.1),

$$\Pr(A) = \Pr(AB + AB') = \Pr(AB) + \Pr(AB') \tag{1.1.4.4}$$

Rearranging, we get:

$$P(AB') = P(A) - P(AB) \tag{1.1.4.5}$$

1.1.5 The Laplace transform is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{1.1.5.1}$$

1.1.6

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{1.1.6.1}$$

Solution: Applying integration by parts,

$$u = e^{-st}, dv = g'(t)\, dt \tag{1.1.6.2}$$

$$\implies \int_0^\infty e^{-st} g'(t)\, dt = \left[e^{-st} g(t)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) g(t)\, dt \tag{1.1.6.3}$$

$$= -g(0) + s\int_0^\infty e^{-st} g(t)\, dt \tag{1.1.6.4}$$

yielding (1.1.6.1).

1.1.7

$$g''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2 G(s) - sg(0) - g'(0) \tag{1.1.7.1}$$

1.1.8

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.8.1}$$

Solution:

$$\implies e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{-st} \left(e^{at} g(t)\right) dt \tag{1.1.8.2}$$

$$= \int_0^\infty e^{-(s-a)t} g(t)\, dt \tag{1.1.8.3}$$

yielding (1.1.8.1).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \tag{1.1.9.1}$$

1.1.10

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \tag{1.1.10.1}$$

Solution:

$$\int_0^\infty e^{-st}\, dt = \left(-\frac{e^{-st}}{s}\right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \tag{1.1.10.2}$$

1.1.11

$$tu(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s^2} \tag{1.1.11.1}$$

Solution: Differentiating (1.1.10.2) with respect to s results in (1.1.11.1).

1.1.12

$$e^{-at} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+a}, \quad \Re(s) > \Re(a). \tag{1.1.12.1}$$

Solution: See (1.1.10.1) and (1.1.8.1).

1.1.13

$$\begin{aligned} \cos(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2} \\ \sin(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \end{aligned} \quad \Re(s) > a \tag{1.1.13.1}$$

Solution: From (1.1.8.1) and (1.1.10.1),

$$e^{\jmath at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-\jmath a}, \quad \Re(s) > a \tag{1.1.13.2}$$

$$\implies \cos(at)u(t) + \jmath \sin(at)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s+\jmath a}{(s-\jmath a)(s+\jmath a)} \tag{1.1.13.3}$$

$$= \frac{s}{s^2+a^2} + \jmath \frac{a}{s^2+a^2} \tag{1.1.13.4}$$

resulting in (1.1.13.1).

1.1.14 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \tag{1.1.14.1}$$

Solution: Applying the Laplace transform on both sides, and using (1.1.7.1),

$$s^2Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \tag{1.1.14.2}$$

Substituting

$$y(0) = \beta, y'(0) = \gamma \tag{1.1.14.3}$$

in (1.1.14.2),

$$Y(s) = \frac{s\beta + \gamma}{s^2+\alpha^2} \tag{1.1.14.4}$$

$$= \frac{s\beta}{s^2+\alpha^2} + \frac{\gamma}{s^2+\alpha^2} \tag{1.1.14.5}$$

yielding

$$y(t) = \beta\cos(\alpha t) + \frac{\gamma}{\alpha}\sin(\alpha t) \tag{1.1.14.6}$$

from (1.1.13.1).

1.1.15 (Euler's Method of Forward Differences.) Numerically solve

$$\frac{dy}{dx} = f(x,y), y(x_0) = y_0 \tag{1.1.15.1}$$

Solution: By definition,

$$\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{h\to 0} \frac{y(x_0+h) - y(x_0)}{h} \tag{1.1.15.2}$$

which can be expressed using the notation

$$y_0' = \frac{y_1 - y_0}{h} \tag{1.1.15.3}$$

Similarly,

$$y_n'' = \frac{y_{n+1}' - y_n'}{h} \tag{1.1.15.4}$$

From (1.1.15.3),

$$y'_{n+1} = \frac{y_{n+2} - y_{n+1}}{h}, y'_n = \frac{y_{n+1} - y_n}{h} \tag{1.1.15.5}$$

yielding

$$y''_n = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.15.6}$$

upon simplification.

1.1.16 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.16.1}$$

1.1.17

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi n f_0 t}\, dt = \delta_n \tag{1.1.17.1}$$

1.1.18 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)\,(t - nT_0) \tag{1.1.18.1}$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \tag{1.1.18.2}$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \tag{1.1.18.3}$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.18.4}$$

Solution: From (1.1.18.3),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \right) e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.18.5}$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \tag{1.1.18.6}$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \tag{1.1.18.7}$$

upon substituting from (1.1.18.2) and (1.1.17.1), yielding (1.1.18.4).

1.1.19 The rectangular function. The unit rectangular function is defined as

$$\text{rect}\,(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{1.1.19.1}$$

1.1.20 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}\,(t), \quad T_0 = 2. \tag{1.1.20.1}$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \tag{1.1.20.2}$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (1.1.18.4),

$$c_n = \frac{1}{2}\int_{-1}^{1} e^{-j\pi kt}\, dt \tag{1.1.20.3}$$

$$= \left[\frac{\sin(\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{\frac{1}{2}} \tag{1.1.20.4}$$

The following code verifies the result in Fig. 1.1.20.1. Note that Fig. 1.1.20.1 is continuous while (1.1.20.1) is discontinuous. So (1.1.20.2) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

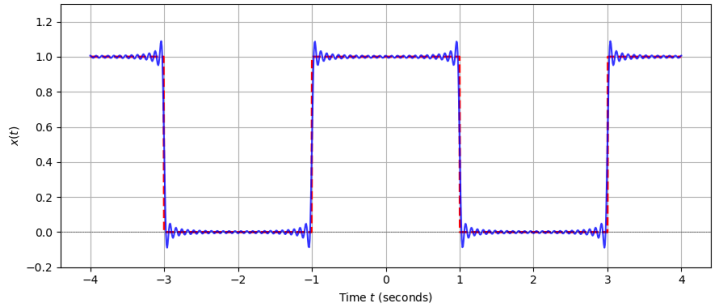


Fig. 1.1.20.1. Gibbs phenomenon

1.1.21 If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} c_n, \tag{1.1.21.1}$$

show that

$$x(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} c_n e^{-\mathrm{J}2\pi n t_0}, \tag{1.1.21.2}$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t-t_0\right) e^{-\jmath 2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t\right) e^{-\jmath 2\pi n f_0 (t+t_0)}\, dt \tag{1.1.21.3}$$

$$= e^{\jmath 2\pi n f_0 t_0} \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0\left(t\right) e^{-\jmath 2\pi n f_0 t}\, dt \tag{1.1.21.4}$$

after a change of variables. Note that the limits of integration are unchanged due the periodicity of $x(t)$. Transferring the exponential term in (1.1.21.4) to the L.H.S yields (1.1.21.2).

1.1.22 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.1.22.1}$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.1.22.2}$$

1.1.23 Find the echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.23.1}$$

Solution: Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.1.23.2}$$

1.1.24 The rank of a matrix is defined to be the number of nonzero rows in echelon form. $\therefore \text{rank}\left(\mathbf{A}\right) = 1$.

1.1.25 A matrix is said to be singular if its rank is nonzero.

1.1.26 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}\left(\mathbf{A}\right) = 1. \tag{1.1.26.1}$$

where both $\mathbf{u}$ and $\mathbf{v}$ are n-dimensional real-valued non-zero column vectors.

Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \tag{1.1.26.2}$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.1.26.3}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.26.4}$$

From (1.1.23.2) and Definition 1.1.24, we obtain (1.1.26.1).

1.1.27

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right| \tag{1.1.27.1}$$

1.1.28

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.28.1}$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k \tag{1.1.28.2}$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to verify that

$$\left|\mathbf{EB}\right| = \left|\mathbf{E}\right| \left|\mathbf{B}\right|. \tag{1.1.28.3}$$

$$\therefore \left|\mathbf{AB}\right| = \left|\mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k\mathbf{B}\right| \tag{1.1.28.4}$$

$$= \left|\mathbf{E}_1\right| \left|\mathbf{E}_2 \cdots \mathbf{E}_k\mathbf{B}\right| \tag{1.1.28.5}$$

$$= \left|\mathbf{E}_1\right| \left|\mathbf{E}_2\right| \cdots \left|\mathbf{E}_k\right| \left|\mathbf{B}\right| \tag{1.1.28.6}$$

For $\mathbf{B} = \mathbf{I}$,

$$\left|\mathbf{A}\right| = \left|\mathbf{E}_1\right| \left|\mathbf{E}_2\right| \cdots \left|\mathbf{E}_k\right| \tag{1.1.28.7}$$

which upon substituting in (1.1.28.6) yields

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.28.8}$$

1.1.29

$$\left|\mathbf{A^{-1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.29.1}$$

1.1.30 If

$$\mathbf{Ax} = \lambda\mathbf{x}, \tag{1.1.30.1}$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.31 Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^{n} \lambda_i = \text{Tr}\left(\mathbf{A}\right) \tag{1.1.31.1}$$

$$\prod_{i=1}^{n} \lambda_i = \left|\mathbf{A}\right| \tag{1.1.31.2}$$

1.1.32 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.
Solution: From (1.1.29.1),

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = \left|\left(\mathbf{A} - \lambda\mathbf{I}\right)^\top\right| \tag{1.1.32.1}$$

$$= \left|\mathbf{A}^\top - \lambda\mathbf{I}\right|, \tag{1.1.32.2}$$

completing the proof.

1.1.33 The equation of a conic with directrix $\mathbf{n}^\top\mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\text{g}\left(\mathbf{x}\right) = \mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0 \tag{1.1.33.1}$$

where

$$\mathbf{V} = \left\|\mathbf{n}\right\|^2\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.33.2}$$

$$\mathbf{u} = ce^2\mathbf{n} - \left\|\mathbf{n}\right\|^2\mathbf{F}, \tag{1.1.33.3}$$

$$f = \left\|\mathbf{n}\right\|^2\left\|\mathbf{F}\right\|^2 - c^2e^2 \tag{1.1.33.4}$$

Solution:

$$\left\|\mathbf{x} - \mathbf{F}\right\|^2 = e^2\frac{\left(\mathbf{n}^\top\mathbf{x} - c\right)^2}{\left\|\mathbf{n}\right\|^2} \tag{1.1.33.5}$$

$$\implies \left\|\mathbf{n}\right\|^2\left(\mathbf{x} - \mathbf{F}\right)^\top\left(\mathbf{x} - \mathbf{F}\right) = e^2\left(\mathbf{n}^\top\mathbf{x} - c\right)^2 \tag{1.1.33.6}$$

$$\implies \left\|\mathbf{n}\right\|^2\left(\mathbf{x}^\top\mathbf{x} - 2\mathbf{F}^\top\mathbf{x} + \left\|\mathbf{F}\right\|^2\right) = e^2\left(c^2 + \left(\mathbf{n}^\top\mathbf{x}\right)^2 - 2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.33.7}$$

$$= e^2\left(c^2 + \left(\mathbf{x}^\top\mathbf{n}\mathbf{n}^\top\mathbf{x}\right) - 2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.33.8}$$

which can be expressed as (1.1.33.1) after simplification.

1.1.34 The polar form of a conic section is given as:

$$r = \frac{ep}{1 + e\cos\left(\theta - \theta_0\right)} \tag{1.1.34.1}$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (1.1.33.1), the general equation of a conic in matrix form is given

by

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1.1.34.2}$$

For a conic with a focus at the origin ($\mathbf{F} = \mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix}\cos\theta_0 \\ \sin\theta_0\end{pmatrix}$, eccentricity e, and directrix distance parameter $c = -p$, substituting into (1.1.33.2)–(1.1.33.4) yields:

$$\mathbf{V} = \mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.34.3}$$

$$\mathbf{u} = -pe^2\mathbf{n}, \tag{1.1.34.4}$$

$$f = -p^2e^2 \tag{1.1.34.5}$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r\mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} \tag{1.1.34.6}$$

Substituting into (1.1.34.2) yields

$$r^2\left(\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta)\right) + 2r\left(\mathbf{u}^\top \mathbf{r}(\theta)\right) + f = 0 \tag{1.1.34.7}$$

Evaluating the inner products:

$$\mathbf{n}^\top \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta_0 & \sin\theta_0\end{pmatrix}\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} = \cos(\theta - \theta_0) \tag{1.1.34.8}$$

$$\mathbf{r}(\theta)^\top \mathbf{V}\mathbf{r}(\theta) = \mathbf{r}(\theta)^\top \left(\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top\right)\mathbf{r}(\theta) = 1 - e^2\cos^2(\theta - \theta_0) \tag{1.1.34.9}$$

$$\mathbf{u}^\top \mathbf{r}(\theta) = -pe^2\cos(\theta - \theta_0) \tag{1.1.34.10}$$

Substituting these into (1.1.34.7):

$$r^2\left[1 - e^2\cos^2(\theta - \theta_0)\right] - 2rpe^2\cos(\theta - \theta_0) - p^2e^2 = 0 \tag{1.1.34.11}$$

Factoring the quadratic expression:

$$\left[r\left(1 + e\cos(\theta - \theta_0)\right) - ep\right]\left[r\left(1 - e\cos(\theta - \theta_0)\right) + ep\right] = 0 \tag{1.1.34.12}$$

Taking the positive real root for radial distance $r > 0$:

$$r\left(1 + e\cos(\theta - \theta_0)\right) = ep \tag{1.1.34.13}$$

yielding (1.1.34.1)

1.1.35 Gravitational parameter of a body μ is defined as

$$\mu = GM \tag{1.1.35.1}$$

where M is the mass of the body

1.1.36 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h = \|\mathbf{r} \times \mathbf{v}\| \tag{1.1.36.1}$$

1.1.37 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} \tag{1.1.37.1}$$

where

$$r = \|\mathbf{r}\|, v = \|\mathbf{v}\| \tag{1.1.37.2}$$

1.1.38 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a} = -\frac{\mu}{r^3}\mathbf{r} \tag{1.1.38.1}$$

1.1.39 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r} \times \mathbf{a} = \mathbf{r} \times \left(-\frac{\mu}{r^3}\mathbf{r}\right) = \mathbf{0} \tag{1.1.39.1}$$

From (1.1.39.1)

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{0} \tag{1.1.39.2}$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{constant} \tag{1.1.39.3}$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right) \times (\mathbf{r} \times \mathbf{v}) \tag{1.1.39.4}$$

$$= -\frac{\mu}{r^3}\left[(\mathbf{r} \cdot \mathbf{v})\,\mathbf{r} - (\mathbf{r} \cdot \mathbf{r})\,\mathbf{v}\right] \tag{1.1.39.5}$$

$$= \frac{\mu}{r^3}\left[r^2\mathbf{v} - (\mathbf{r} \cdot \mathbf{v})\,\mathbf{r}\right] \tag{1.1.39.6}$$

Using $\mathbf{r} \cdot \mathbf{v} = r\frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu\left[\frac{\mathbf{v}}{r} - \frac{1}{r^2}\frac{dr}{dt}\mathbf{r}\right] = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right) \tag{1.1.39.7}$$

Integrating with respect to time t:

$$\mathbf{v} \times \mathbf{h} = \mu\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.39.8}$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (1.1.39.8) with $\mathbf{r}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = \mu\mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.39.9}$$

$$\implies (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mu\left(r + rC\cos\theta\right) \tag{1.1.39.10}$$

$$\implies \|\mathbf{h}\|^2 = \mu r\left(1 + C\cos\theta\right) \tag{1.1.39.11}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C\cos\theta} = \frac{ep}{1 + e\cos\theta} \tag{1.1.39.12}$$

This is the same equation as (1.1.34.1) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \tag{1.1.39.13}$$

$$\mathbf{e} = \mathbf{C} \tag{1.1.39.14}$$

1.1.40 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.40.1}$$

Solution: From (1.1.39.8) and (1.1.39.14),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \tag{1.1.40.2}$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \tag{1.1.40.3}$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v} \times \mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r}\left[\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h})\right] + \frac{\mathbf{r} \cdot \mathbf{r}}{r^2} \tag{1.1.40.4}$$

We know that

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mathbf{h} \cdot \mathbf{h} = h^2 \tag{1.1.40.5}$$

And,

$$\mathbf{r} \cdot \mathbf{r} = r^2 \tag{1.1.40.6}$$

Substituting these back into (1.1.40.4) yields

$$e^2 = \frac{v^2 h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \tag{1.1.40.7}$$

$$= 1 + \frac{2h^2}{\mu^2}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) \tag{1.1.40.8}$$

Substituting (1.1.37.1) into (1.1.40.8) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.40.9}$$

1.1.41 Consider the system A system comprising a bar, spring and mass is shown in the

figure Fig. 1.1.41.1 The bar, having negligible mass, is made of a material having Young's modulus E, cross-sectional area A and length L. The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

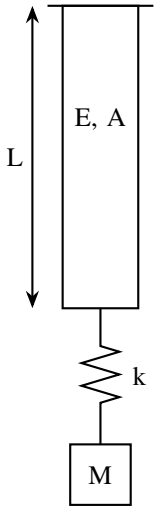


Fig. 1.1.41.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \tag{1.1.41.1}$$

1.1.42 Find the overall spring constant in Fig. 1.1.41.1.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \tag{1.1.42.1}$$

$$\implies k_{eq} = \frac{k_b k}{k_b + k} \tag{1.1.42.2}$$

where k_t is the overall spring constant.

1.1.43 Show that the equation for motion for the mass in Fig. 1.1.43.1 is

$$\frac{d^2x}{dt^2} = -\frac{k}{M}x \tag{1.1.43.1}$$

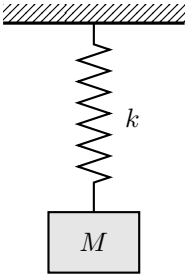


Fig. 1.1.43.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st}.Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \tag{1.1.43.2}$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M\frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \tag{1.1.43.3}$$

$$\tag{1.1.43.4}$$

yielding (1.1.43.1), upon substituting from (1.1.43.2).

2. BT 2025

A. Formulae

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.1.1.1}$$

2.1.2 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.2.1}$$

2.1.3 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \tag{2.1.3.1}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0 \tag{2.1.3.2}$$

2.1.4 A circle with center $\mathbf{c}$ and radius r, can be expressed as (1.1.33.1) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{2.1.4.1}$$

Solution: The equation of a circle is

$$\|\mathbf{x}-\mathbf{c}\|^2 = r^2 \tag{2.1.4.2}$$

$$(\mathbf{x}-\mathbf{c})^\top (\mathbf{x}-\mathbf{c}) = r^2 \tag{2.1.4.3}$$

$$\mathbf{x}^\top\mathbf{x} - 2\mathbf{c}^\top\mathbf{x} + \mathbf{c}^\top\mathbf{c} - r^2 = 0 \tag{2.1.4.4}$$

yielding (2.1.4.1) upon comparing with (1.1.33.1)

2.1.5 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h}+\kappa\mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0$ are given by

$$\mathbf{X} = \begin{pmatrix}\mathbf{h} & \mathbf{m}\end{pmatrix}\begin{pmatrix}1 & 1\\ \kappa_1 & \kappa_2\end{pmatrix} \tag{2.1.5.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top\mathbf{V}\mathbf{m})}}{\mathbf{m}^\top\mathbf{V}\mathbf{m}}, i = 1, 2. \tag{2.1.5.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top\mathbf{V}\mathbf{h} + 2\mathbf{u}^\top\mathbf{h} + f$.

2.1.6 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil

$$\mathbf{x}^\top(\mathbf{V}_1 + \mu\mathbf{V}_2)\mathbf{x} + 2(\mathbf{u}_1 + \mu\mathbf{u}_2)^\top\mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.1.6.1}$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\begin{vmatrix}\mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2\\ (\mathbf{u}_1 + \mu\mathbf{u}_2)^\top & f_1 + \mu f_2\end{vmatrix} = 0 \tag{2.1.6.2}$$

2.1.7 For a differentiable function $f(x)$,

$$f''(x) > 0 \quad \forall x \in (0,\infty) \implies f(x) \text{ is strictly convex} \tag{2.1.7.1}$$

2.1.8 (Gradient Descent Optimization) For the convex optimization problem

$$\min_x f(x), \tag{2.1.8.1}$$

$$x_{k+1} = x_k - \eta f'(x_k) \tag{2.1.8.2}$$

is the iterative parameter update with learning rate $\eta > 0$.

2.1.9 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{2.1.9.1}$$

2.1.10 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.10.1}$$

is consistent iff

$$\text{rank}\left(\mathbf{A}\right) = \text{rank}\left(\left[\mathbf{A} \mid \mathbf{b}\right]\right) = r \tag{2.1.10.2}$$

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1,2,3,4,5,6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF)

$$p_{X_1}\left(n\right) = p_{X_2}\left(n\right) = \begin{cases} \frac{1}{6}, & 1 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.11.1}$$

- (Convolution) Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}\left(n\right)$ and $p_{X_2}\left(n\right)$:

$$p_X\left(n\right) = \Pr\left(X_1 + X_2 = n\right) = \sum_k \Pr\left(X_1 = n-k \mid X_2 = k\right) p_{X_2}\left(k\right) \tag{2.1.11.2}$$

$$= \sum_k p_{X_1}\left(n-k\right) p_{X_2}\left(k\right) = p_{X_1}\left(n\right) * p_{X_2}\left(n\right) \tag{2.1.11.3}$$

- Substituting (2.1.11.1) into (2.1.11.3) yields the triangular PMF

$$p_X\left(n\right) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \leq n \leq 7 \\ \frac{13-n}{36}, & 7 < n \leq 12 \\ 0, & n > 12 \end{cases} \tag{2.1.11.4}$$

The following code verifies (2.1.11.4) in Fig. 2.1.11.1.

codes/btq35/dice.py

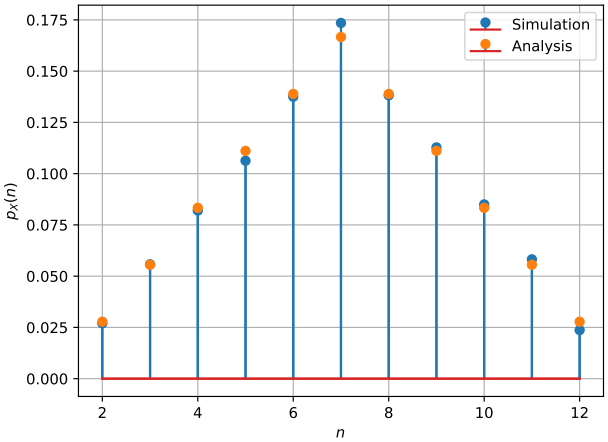


Fig. 2.1.11.1. PMF of the sum of two fair dice: Simulation vs Analysis

2.1.12 (Unit Step Response of an RC Circuit) Consider a series RC circuit excited by a step input $v_i(t) = V_0 u(t)$ where V_0 is a constant. Applying KVL around the loop,

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.1}$$

yielding

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.2}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.3}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.4}$$

$$\implies v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.5}$$

2.1.13 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ iff

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.13.1}$$

2.1.14 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h\to 0^-}\frac{f(x_0+h)-f(x_0)}{h} = \lim_{h\to 0^+}\frac{f(x_0+h)-f(x_0)}{h} = f'\left(x_0^+\right) \tag{2.1.14.1}$$

2.1.15 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \tag{2.1.15.1}$$

2.1.16 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \tag{2.1.16.1}$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.16.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f_1'(a) = f_2'(a) \tag{2.1.16.3}$$

2.1.17 The Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \tag{2.1.17.1}$$

over the Region of Convergence (ROC) where the series converges absolutely.

2.1.18 The discrete unit step function is defined as

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.1.18.1}$$

and

$$u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \tag{2.1.18.2}$$

2.1.19 Time-Shift Property

$$x(n - k) \overset{\mathcal{Z}}{\longleftrightarrow} z^{-k} X(z) \tag{2.1.19.1}$$

2.1.20 Standard Inverse Transform Pair: For any scalar constant α,

$$\alpha^n u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - \alpha z^{-1}}, \quad |z| > |\alpha| \tag{2.1.20.1}$$

2.1.21 (Trace of a Matrix) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix with eigenvalues

$\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}\,(\mathbf{A}) \tag{2.1.21.1}$$

Solution: The characteristic polynomial of $\mathbf{A}$ is defined by

$$p_{\mathbf{A}}(\lambda) = \left|\lambda\mathbf{I} - \mathbf{A}\right| \tag{2.1.21.2}$$

$$= \prod_{i=1}^{n} (\lambda - \lambda_i) = \lambda^n - \left(\sum_{i=1}^{n} \lambda_i\right) \lambda^{n-1} + \cdots + (-1)^n \prod_{i=1}^{n} \lambda_i \tag{2.1.21.3}$$

Evaluating the determinant $\left|\lambda\mathbf{I} - \mathbf{A}\right|$ using cofactor expansion, the only term that contributes to λ^n and λ^{n-1} is the product of the principal diagonal entries

$$\prod_{i=1}^{n} (\lambda - a_{ii}) = \lambda^n - \left(\sum_{i=1}^{n} a_{ii}\right) \lambda^{n-1} + \ldots \tag{2.1.21.4}$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$. Equating the coefficient of λ^{n-1} from (2.1.21.3) and (2.1.21.4)

3. AG 2025

A. Formulae

3.1.1 (Mean) The mean of a random variable X with probability mass function $p_X(k)$ is defined as

$$E[X] = \sum_{k} x_k p_X(k) \tag{3.1.1.1}$$

3.1.2 (Bernoulli Distribution) For $X \in \{0, 1\}$, the pmf is defined as

$$P_X(k) = \begin{cases} p & k = 1 \\ q & k = 0 \\ 0 & \text{otherwise} \end{cases} \tag{3.1.2.1}$$

3.1.3 The moment generating function

$$\begin{aligned} M_X(z) \triangleq E\left[z^{-X}\right] &= p_X(1) z^{-1} + p_X(0) z^0 \\ &= p z^{-1} + q \end{aligned} \tag{3.1.3.1}$$

upon substituting from (2.1.17.1).

3.1.4 (Binomial Distribution) If X_i are independent, for

$$\begin{aligned} Y \triangleq \sum_{i=1}^{n} X_i, \quad M_Y(z) &= \prod_{i=1}^{n} M_{X_i}(z) \\ &= \left(pZ^{-1} + q\right)^n \end{aligned} \tag{3.1.4.1}$$

Solution: From (3.1.3.1),

$$M_Y(z) = E\left[z^{-\sum_{i=1}^{n} X_i}\right] = E\left[\prod_{i=1}^{n} z^{-X_i}\right] \tag{3.1.4.2}$$

$$= \prod_{i=1}^{n} E\left[z^{-X_i}\right] \tag{3.1.4.3}$$

yielding (3.1.4.1).

3.1.5 From (3.1.4.1),

$$\left(pz^{-1} + q\right)^n = {}^nC_k p^k q^{n-k} z^{-k} \tag{3.1.5.1}$$

$$\implies p_Y(k) = \begin{cases} {}^nC_k p^k q^{n-k} & 0 \le k \le n \\ 0 & \text{otherwise} \end{cases} \tag{3.1.5.2}$$

using the definition in (3.1.3.1) along with (2.1.17.1).

3.1.6

$$\begin{aligned} \sinh(at)\, u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2 - a^2} \\ \cosh(at)\, u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2 - a^2} \end{aligned} \tag{3.1.6.1}$$

Solution: From (1.1.13.1),

$$\cos(\jmath at) u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2 + (\jmath a)^2} \tag{3.1.6.2}$$

yielding (3.1.6.1).

3.1.7 (Geometric Progression)

$$\sum_{i=0}^{n-1} q^i = \frac{q^n - 1}{q - 1} \tag{3.1.7.1}$$

3.1.8

$$\begin{aligned} \sin(x) &= \sum_{i=1}^{n} -1^{i+1} \frac{x^{2i-1}}{(2i-1)!} \\ \cos(x) &= \sum_{i=1}^{n} -1^{i+1} \frac{x^{i-1}}{i-1!} \end{aligned} \tag{3.1.8.1}$$

3.1.9 If

$$f(\alpha x, \alpha y) = \alpha^n \alpha^n f(x, y), \tag{3.1.9.1}$$

$f(x, y)$ is homogeneous of degree n. Then,

$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = nf \tag{3.1.9.2}$$

4. GATE CE-2 2025

A. Formulae

4.1.1 (Integration by Parts) For differentiable functions $u(x)$ and $v(x)$, the integration by parts formula is:

$$\int u\,dv = uv - \int v\,du \tag{4.1.1.1}$$

4.1.2 Let

$$\mathbf{V} = \begin{pmatrix} u \\ v \end{pmatrix}, \nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}, \tag{4.1.2.1}$$

$$\tag{4.1.2.2}$$

The curl and divergence of $\mathbf{V}$ are then defined respectively as

$$\nabla \times \mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{e}_3 \tag{4.1.2.3}$$

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \tag{4.1.2.4}$$

4.1.3 If $B \subset A, AB = 1$.

4.1.4 (Conditional Probability)

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \quad \Pr(B) > 0. \tag{4.1.4.1}$$

4.1.5 (Variance)

$$\sigma^2 = E\left[X^2\right] - (E[X])^2 \tag{4.1.5.1}$$

5. GATE-BM-2025

A. Formulae

5.1.1 Geometric Progression of Frequencies

If each successive term is r times the previous term, then after k steps from a reference term f_0,

$$\frac{f_k}{f_0} = r^k. \tag{5.1.1.1}$$

5.1.2 Self-Similar (Fractal) Curve Length

If, at every iteration, each segment is replaced by m segments each of length $\frac{1}{p}$ of the original segment, the length scales as

$$L_1 = L_0 \left(\frac{m}{p} \right), \tag{5.1.2.1}$$

$$L_n = L_0 \left(\frac{m}{p} \right)^n. \tag{5.1.2.2}$$

5.1.3 Notch (Band-Stop) Filter Response

A notch filter strongly attenuates a narrow band of frequencies around a rejected frequency ω_0, while passing frequencies away from it:

$$|H(j\omega_0)| \approx 0, \tag{5.1.3.1}$$

$$|H(j\omega)| \text{ high for } \omega < \omega_0 \text{ and } \omega > \omega_0. \tag{5.1.3.2}$$

5.1.4 Rolle's Theorem

If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then

$$\exists\, c \in (a, b) \text{ such that } f'(c) = 0. \tag{5.1.4.1}$$

Contrapositive form: if $f'(x) \neq 0$ for all x, then $f(a) \neq f(b)$ for $a \neq b$.

5.1.5 Standard Second-Order System / Closed-Loop Transfer Function

For unity negative feedback with forward transfer function $G(s)$,

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}. \tag{5.1.5.1}$$

Comparing the characteristic equation with the standard second-order form,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \tag{5.1.5.2}$$

gives the natural frequency ω_n and damping ratio ζ. The system is:

- overdamped if $\zeta > 1$,
- critically damped if $\zeta = 1$,
- underdamped if $0 < \zeta < 1$,
- undamped (sustained oscillations) if $\zeta = 0$.

For $0 < \zeta < 1$, the impulse response has the form

$$c(t) = A\, e^{-\zeta\omega_n t} \sin\!\left(\omega_n \sqrt{1 - \zeta^2}\, t\right), \qquad t \geq 0. \tag{5.1.5.3}$$

5.1.6 Eigenvalues via the Characteristic Equation

For a matrix $\mathbf{A}$, the eigenvalues λ satisfy

$$\det(A - \lambda I) = 0. \tag{5.1.6.1}$$

For a 2×2 rotation matrix

$$\mathbf{A} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{5.1.6.2}$$

Therefore,
Expanding the determinant,

$$(\cos\theta - \lambda)^2 + \sin^2\theta = 0. \tag{5.1.6.3}$$

Expanding further,

$$\cos^2\theta - 2\lambda\cos\theta + \lambda^2 + \sin^2\theta = 0. \tag{5.1.6.4}$$

Using

$$\sin^2\theta + \cos^2\theta = 1, \tag{5.1.6.5}$$

we get

$$\lambda^2 - 2\lambda\cos\theta + 1 = 0. \tag{5.1.6.6}$$

Using the quadratic formula,

$$\lambda = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}. \tag{5.1.6.7}$$

Since

$$4\cos^2\theta - 4 = -4\sin^2\theta, \tag{5.1.6.8}$$

we have

$$\lambda = \frac{2\cos\theta \pm 2i\sin\theta}{2}. \tag{5.1.6.9}$$

Hence,

$$\lambda = \cos\theta \pm i\sin\theta. \tag{5.1.6.10}$$

Using Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{5.1.6.11}$$

and

$$e^{-i\theta} = \cos\theta - i\sin\theta, \tag{5.1.6.12}$$

using $\sin^2\theta + \cos^2\theta = 1$ and Euler's formula $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$, the eigenvalues reduce to

$$\lambda = \cos\theta \pm i\sin\theta = e^{\pm i\theta}. \tag{5.1.6.13}$$

5.1.7 Definition of the Derivative (First Principles)

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}. \tag{5.1.7.1}$$

5.1.8 Mean and Median of a Dataset

For n ordered values $x_1 \le x_2 \le \cdots \le x_n$:

$$\text{Mean} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{5.1.8.1}$$

If n is odd, the median is the middle term,

$$\text{Median} = x_{\frac{n+1}{2}}. \tag{5.1.8.2}$$

5.1.9 Nyquist Sampling Theorem

A band-limited signal with maximum frequency component $f_{\max}$ must be sampled at a rate

$$f_s \ge 2f_{\max} \tag{5.1.9.1}$$

to allow accurate (aliasing-free) reconstruction. For an angular frequency component ω, the corresponding cyclic frequency is

$$f = \frac{\omega}{2\pi}. \tag{5.1.9.2}$$

5.1.10 Average Power in an AC Circuit

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\theta, \tag{5.1.10.1}$$

where

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \tag{5.1.10.2}$$

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}} \tag{5.1.10.3}$$

$$\theta = \phi_v - \phi_i. \tag{5.1.10.4}$$

5.1.11 Mean and Variance of a Binomial Distribution
For $X \sim B(N, p)$,

$$E(X) = Np \tag{5.1.11.1}$$

$$\text{Var}(X) = Np(1-p). \tag{5.1.11.2}$$

5.1.12 Solving a Linear ODE via the Laplace Transform
For $y''(x) - k^2 y(x) = 0$ with initial conditions $y(0)$, $y'(0)$, take the Laplace transform using

$$\mathcal{L}\{y''(x)\} = s^2 Y(s) - s\,y(0) - y'(0). \tag{5.1.12.1}$$

This gives an algebraic equation for $Y(s)$, which is solved by partial fractions,

$$Y(s) = \frac{A}{s-a} + \frac{B}{s+a}, \tag{5.1.12.2}$$

and inverted using

$$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}. \tag{5.1.12.3}$$

5.1.13 Impedance of R–C Networks (Phasor Domain)
Symbolic Impedance Derivation
The impedance of the capacitor in terms of angular frequency ω is:

$$Z_C(\omega) = \frac{1}{j\omega C} = -\frac{j}{\omega C} \tag{5.1.13.1}$$

The parallel combination of resistor R_2 and capacitor $Z_C(\omega)$ gives $Z_p(\omega)$:

$$Z_p(\omega) = \frac{R_2 \cdot Z_C(\omega)}{R_2 + Z_C(\omega)} \tag{5.1.13.2}$$

$$= \frac{R_2\left(-\frac{j}{\omega C}\right)}{R_2 - \frac{j}{\omega C}} \tag{5.1.13.3}$$

$$= \frac{-jR_2}{\omega R_2 C - j} \tag{5.1.13.4}$$

Multiplying numerator and denominator by the complex conjugate $(\omega R_2 C + j)$:

$$Z_p(\omega) = \frac{-jR_2(\omega R_2 C + j)}{(\omega R_2 C - j)(\omega R_2 C + j)} \tag{5.1.13.5}$$

$$= \frac{R_2 - j\omega R_2^2 C}{1 + (\omega R_2 C)^2} \tag{5.1.13.6}$$

The total input impedance $Z_{\text{in}}(\omega)$ in series with R_1 is:

$$Z_{\text{in}}(\omega) = R_1 + Z_p(\omega) \tag{5.1.13.7}$$

$$= \left(R_1 + \frac{R_2}{1 + (\omega R_2 C)^2}\right) - j\left(\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2}\right) \tag{5.1.13.8}$$

Magnitude and Phase Expressions in terms of ω
1. Magnitude Spectrum $|Z_{\text{in}}(\omega)|$:

$$|Z_{\text{in}}(\omega)| = \sqrt{\left[R_1 + \frac{R_2}{1 + (\omega R_2 C)^2}\right]^2 + \left[\frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2}\right]^2} \tag{5.1.13.9}$$

Simplifying the algebraic expression:

$$|Z_{\text{in}}(\omega)| = \sqrt{R_1^2 + \frac{2R_1 R_2 + R_2^2 + (\omega R_1 R_2 C)^2}{1 + (\omega R_2 C)^2}} \tag{5.1.13.10}$$

2. Phase Spectrum $\angle Z_{\text{in}}(\omega)$:

$$\angle Z_{\text{in}}(\omega) = -\tan^{-1}\left(\frac{\frac{\omega R_2^2 C}{1+(\omega R_2 C)^2}}{R_1 + \frac{R_2}{1+(\omega R_2 C)^2}}\right) \tag{5.1.13.11}$$

$$= -\tan^{-1}\left(\frac{\omega R_2^2 C}{R_1(1 + (\omega R_2 C)^2) + R_2}\right) \tag{5.1.13.12}$$

5.1.14 Complex Logarithm

For real x, using Euler's identity for -1,

$$y = \ln(-e^x) \tag{5.1.14.1}$$

Using logarithmic properties, we rewrite the term inside the logarithm:

$$-e^x = (-1) \cdot e^x \tag{5.1.14.2}$$

Applying Euler's identity, -1 can be expressed in complex exponential form as:

$$-1 = e^{i(2k+1)\pi}, \quad \text{for } k \in \mathbb{Z} \tag{5.1.14.3}$$

By solving:

$$-e^{x} = e^{x} \cdot e^{i(2k+1)\pi} \tag{5.1.14.4}$$

$$= e^{x+i(2k+1)\pi} \tag{5.1.14.5}$$

5.1.15 Linearity and Causality of a System

Linearity (superposition): a system is linear if, for inputs x_1, x_2 and scalars a, b,

$$T\{a\,x_1(t) + b\,x_2(t)\} = a\,T\{x_1(t)\} + b\,T\{x_2(t)\}. \tag{5.1.15.1}$$

Causality: a system is causal if the output at time t_0 depends only on input values for $t \le t_0$; if it depends on future input values ($t > t_0$), it is noncausal.

5.1.16 Linear (Discrete) Convolution

For an LTI system with impulse response $h(n)$ of length M and input $x(n)$ of length N,

$$y(n) = h(n) * x(n) \tag{5.1.16.1}$$

$$= \sum_{k} h(k)\,x(n-k), \tag{5.1.16.2}$$

with output length

$$L = M + N - 1. \tag{5.1.16.3}$$

This can be written as a matrix–vector product using a lower triangular Toeplitz matrix $\mathbf{H}$ built from shifted copies of $h(n)$:

$$\mathbf{y} = \mathbf{H}, \mathbf{x}. \tag{5.1.16.4}$$

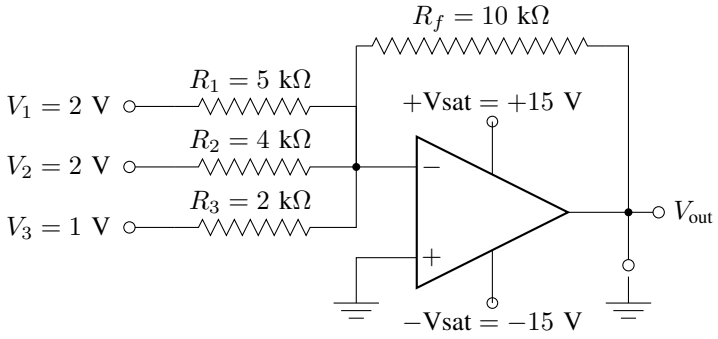
5.1.17 n-bit DAC Output

For an n-bit DAC with full-scale output voltage V_{FS} and digital input equivalent decimal value D,

$$V_{\text{out}} = V_{\text{FS}} \left(\frac{D}{2^{n} - 1} \right). \tag{5.1.17.1}$$

5.1.18 Inverting Summing Amplifier

For an op-amp inverting summing amplifier with feedback resistor R_f, input resistors $R_1, R_2, R_3, \ldots$ and corresponding input voltages $V_1, V_2, V_3, \ldots$, the output voltage is



a) Derivation

Let the inverting input node (where R_1, R_2, R_3, R_f meet) be node A, and let the non-inverting input be grounded.

Ideal op-amp assumptions:

- No current flows into either input terminal.
- The open-loop gain is infinite, so the two inputs are forced to the same potential (virtual short).

Since the non-inverting input is at $0\,\text{V}$ (grounded), the virtual short forces node A to also be at $0\,\text{V}$:

$$V_A = 0 \qquad \text{(virtual ground)}. \tag{5.1.18.1}$$

Applying Kirchhoff's Current Law (KCL) at node A, the sum of currents flowing into the node through R_1, R_2, R_3 must equal the current flowing out through the feedback resistor R_f (since no current enters the op-amp input):

$$\frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} + \frac{V_3 - V_A}{R_3} = \frac{V_A - V_{\text{out}}}{R_f}. \tag{5.1.18.2}$$

Substituting $V_A = 0$:

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_{\text{out}}}{R_f}. \tag{5.1.18.3}$$

Solving for V_{out}:

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) \tag{5.1.18.4}$$

$$= -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right), \tag{5.1.18.5}$$

$$V_{\text{out}} = -\sum_i \frac{R_f}{R_i} V_i. \tag{5.1.18.6}$$

$$V_{\text{out}} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \cdots\right), \tag{5.1.18.7}$$

valid provided V_{out} stays within the saturation limits,

$$-V_{\text{sat}} \le V_{\text{out}} \le +V_{\text{sat}}. \tag{5.1.18.8}$$

5.1.19 Wien Bridge Oscillator Frequency

For a Wien bridge oscillator with equal frequency-selective resistors R and capacitors C, the oscillation frequency is

$$f_0 = \frac{1}{2\pi RC}. \tag{5.1.19.1}$$

5.1.20 Series RLC Circuit — Resonance

Kirchhoff's Voltage Law for a series RLC circuit driven by

$$v(t) = V_m \cos(\omega t) \tag{5.1.20.1}$$

$$R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt = v(t). \tag{5.1.20.2}$$

In the phasor domain, the impedance is

$$Z(\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right) \tag{5.1.20.3}$$

$$|Z(\omega)| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}. \tag{5.1.20.4}$$

Converting to complex phasor notation

$$V(t) = V_{\text{rms}}\sqrt{2}e^{j\omega t} \tag{5.1.20.5}$$

$$I(t) = I_{\text{rms}}\sqrt{2}e^{j(\omega t - \phi)} \tag{5.1.20.6}$$

$$Z(\omega) = \frac{V(\omega)}{I(\omega)} \tag{5.1.20.7}$$

$$= R + j\omega L + \frac{1}{j\omega C} \tag{5.1.20.8}$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right) \tag{5.1.20.9}$$

Resonance occurs when the reactive part vanishes:

$$\omega_0 L = \frac{1}{\omega_0 C} \tag{5.1.20.10}$$

$$\implies \omega_0 = \frac{1}{\sqrt{LC}} \tag{5.1.20.11}$$

$$f_0 = \frac{\omega_0}{2\pi}. \tag{5.1.20.12}$$

At resonance, $Z(\omega_0) = R$, so

$$I_{\text{rms}}(\omega_0) = \frac{V_{\text{rms}}}{R}. \tag{5.1.20.13}$$

5.1.21 Newton–Raphson Method

For a root of $f(x) = 0$, the iterative update starting from x_n is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}. \tag{5.1.21.1}$$

5.1.22 Logarithmic Decay Model

For a quantity that decays logarithmically with time,

$$\sigma(t) = \sigma_0 - k \ln(t), \qquad t \geq 1, \tag{5.1.22.1}$$

the difference between two times t_1, t_2 eliminates the unknown constant σ_0:

$$\sigma(t_2) - \sigma(t_1) = -k \left[\ln(t_2) - \ln(t_1)\right] \tag{5.1.22.2}$$

$$= -k \ln\left(\frac{t_2}{t_1}\right). \tag{5.1.22.3}$$

5.1.23 Gaussian (Normal) Function

A random variable X is said to be Gaussian (normally distributed) with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty. \tag{5.1.23.1}$$

The standard normal form, with $\mu = 0,\ \sigma = 1$, is

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}, \qquad z = \frac{x-\mu}{\sigma}. \tag{5.1.23.2}$$

The function is symmetric about $x = \mu$, and satisfies

$$\int_{-\infty}^{\infty} f(x)\, dx = 1, \qquad E[X] = \mu, \qquad \text{Var}(X) = \sigma^2. \tag{5.1.23.3}$$

a) Binomial Approximation (De Moivre–Laplace Theorem) For a binomial random variable $X \sim \text{Bin}(n, p)$ with large n such that $np \geq 5$ and $n(1-p) \geq 5$, the binomial distribution can be approximated by a Gaussian distribution with:

$$\mu = np, \quad \sigma^2 = np(1-p). \tag{5.1.23.4}$$

The probability mass function is approximated as:

$$\Pr(X = k) \approx \frac{1}{\sqrt{2\pi np(1-p)}} \exp\left(-\frac{(k-np)^2}{2np(1-p)}\right). \tag{5.1.23.5}$$

b) Continuity Correction When approximating a discrete binomial distribution using a continuous Gaussian distribution, a continuity correction factor of 0.5 is applied:

$$\Pr(a \leq X \leq b) \approx P\left(\frac{a-0.5-np}{\sqrt{np(1-p)}} \leq Z \leq \frac{b+0.5-np}{\sqrt{np(1-p)}}\right). \tag{5.1.23.6}$$

5.1.24 Central Limit Theorem (CLT)
Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables, each with finite mean μ and finite variance σ^2 (the underlying distribution need not itself be Gaussian). Define the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{5.1.24.1}$$

Then, as $n \to \infty$, the standardized sample mean converges in distribution to a standard normal random variable:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1). \tag{5.1.24.2}$$

Equivalently, for large n,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \qquad \sum_{i=1}^{n} X_i \approx \mathcal{N}\left(n\mu, n\sigma^2\right). \tag{5.1.24.3}$$

5.1.25 Effect of Scalar Multiplication on a Determinant
For an $n \times n$ matrix A and scalar k,

$$\det(kA) = k^n \det(A). \tag{5.1.25.1}$$

B. Solved Questions

5.2.1 The 12 musical notes are given as $C, C^{\#}, D, D^{\#}, E, F, F^{\#}, G, G^{\#}, A, A^{\#}, B$. Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^{\#}$ and C is: (GATE BM 2025)

(a) $\sqrt[6]{2}$
(b) $\sqrt{2}$
(c) $\sqrt[4]{2}$
(d) 2

Solution:
Each successive musical note has a frequency $\sqrt[12]{2}$ times that of the previous note. From C to $F^{\#}$, there are 6 successive intervals:

$$C \to C^{\#} \to D \to D^{\#} \to E \to F \to F^{\#} \tag{5.2.1.1}$$

From (5.1.1.1)

$$\frac{f_{F\#}}{f_C} = \left(\sqrt[12]{2}\right)^6 \tag{5.2.1.2}$$

$$= 2^{6/12} = 2^{1/2} = \sqrt{2} \tag{5.2.1.3}$$

Hence,

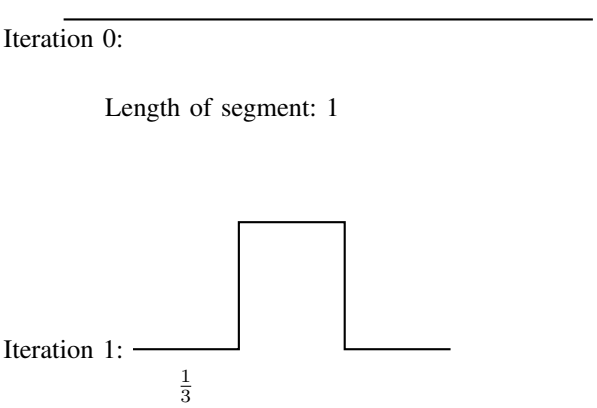
$$\boxed{\frac{f_{F\#}}{f_C} = \sqrt{2}} \tag{5.2.1.4}$$

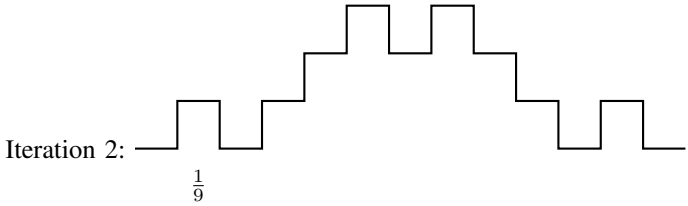
The following code verifies the solution

codes/bm1.py

5.2.2 The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after "Iteration n" is:

GATE BM 2025





Length of each segment: $\frac{1}{9}$

(a) $\left(\frac{5}{3}\right)^{\frac{n}{2}}$
(b) $\left(\frac{5}{3}\right)^{n}$
(c) $\left(\frac{5}{3}\right)^{2n}$
(d) $\left(\frac{5}{3}\right)^{n(2n-1)}$

Solution:
At iteration 0, the length of the curve is 1.

$$L_0 = 1 \tag{5.2.2.1}$$

At iteration 1, the curve consists of 5 segments, each of length $\frac{1}{3}$.
Therefore,

$$L_1 = 5\left(\frac{1}{3}\right) = \frac{5}{3} \tag{5.2.2.2}$$

At every subsequent iteration, each segment is replaced by 5 segments, each having $\frac{1}{3}$ of the length of the original segment. Hence, the total length is multiplied by

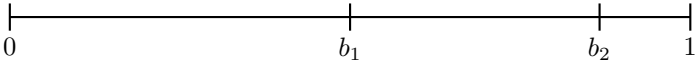
$$\frac{5}{3} \tag{5.2.2.3}$$

Therefore, after n iterations, from (5.1.2.2)

$$L_n = \left(\frac{5}{3}\right)^n \tag{5.2.2.4}$$

5.2.3 A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?
Note: All lengths are in meter. The figure shown is representative.

(GATE BM 2025)



(a) $b_1 < 0.5$
(b) $b_2 > 0.5$
(c) $b_2 < b_1 + 0.5$
(d) $b_1 + b_2 < 1$

Solution:
The stick of length 1 m is broken at b_1 and b_2.
Therefore, the lengths of the three pieces are

$$l_1 = b_1, \qquad l_2 = b_2 - b_1, \qquad l_3 = 1 - b_2. \tag{5.2.3.1}$$

For three lengths to form a triangle, the sum of any two sides must be greater than the third side.
Therefore,

$$l_1 + l_2 > l_3 \tag{5.2.3.2}$$

$$b_1 + (b_2 - b_1) > 1 - b_2 \tag{5.2.3.3}$$

$$b_2 > 1 - b_2 \tag{5.2.3.4}$$

$$b_2 > \frac{1}{2}. \tag{5.2.3.5}$$

Hence, option (B) is a necessary condition.
Now consider

$$l_1 + l_3 > l_2. \tag{5.2.3.6}$$

Substituting the lengths,

$$b_1 + (1 - b_2) > b_2 - b_1. \tag{5.2.3.7}$$

Therefore,

$$2b_1 + 1 > 2b_2 \tag{5.2.3.8}$$

or

$$b_2 < b_1 + \frac{1}{2}. \tag{5.2.3.9}$$

Hence, option (C) is also a necessary condition.
Finally,

$$l_2 + l_3 > l_1. \tag{5.2.3.10}$$

Substituting,

$$(b_2 - b_1) + (1 - b_2) > b_1. \tag{5.2.3.11}$$

Therefore,

$$1 - b_1 > b_1 \tag{5.2.3.12}$$

which gives

$$b_1 < \frac{1}{2}. \tag{5.2.3.13}$$

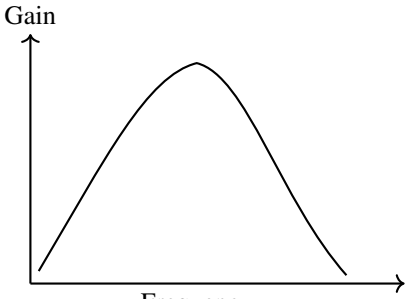
Hence, option (A) is also a necessary condition.
However, the condition

$$b_1 + b_2 < 1 \tag{5.2.3.14}$$

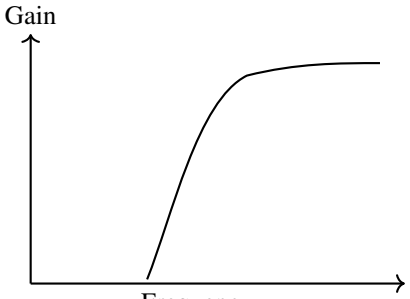
is not required for the three pieces to form a triangle.
Therefore, the condition which is **NOT** necessary is (D)
The following code gives whether the triangle is feasible or not by taking b_1,b_2

Codes/bm3.py

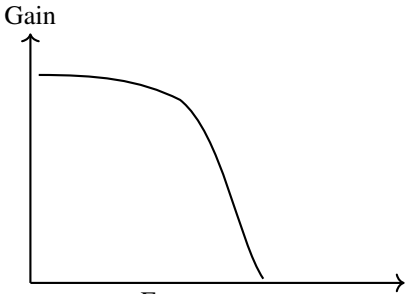
5.2.4 Which one of the following represents the frequency response of a notch filter? (GATE BM 2025)



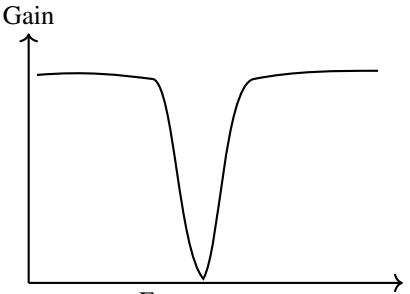
(a)



(b)



(c)



(d)

Solution:
A notch filter is a band-stop filter that rejects or strongly attenuates a narrow range of frequencies while allowing frequencies outside that range to pass.
Therefore, its frequency response has high gain at low and high frequencies, with a sharp reduction in gain at a particular frequency.
The response has a deep attenuation around rejected frequency.
From (5.1.3.2)
Among the given frequency responses, only option (D) shows a sharp reduction in gain at a particular frequency while allowing frequencies on either side to pass.

5.2.5 For any continuous and differentiable function $f(x)$ with first derivative $f'(x) \neq 0$ for all values of x, which one of the following is ALWAYS TRUE for $a \neq b$?
Here, a and b are finite and real.

(GATE BM 2025)

(a) $f(a) > f(b)$
(b) $f(a) = f(b)$
(c) $f(a) \neq f(b)$
(d) $f(a) < f(b)$

Solution:
We are given that $f(x)$ is continuous and differentiable, and

$$f'(x) \neq 0 \qquad \text{for all } x. \tag{5.2.5.1}$$

Suppose, for contradiction, that

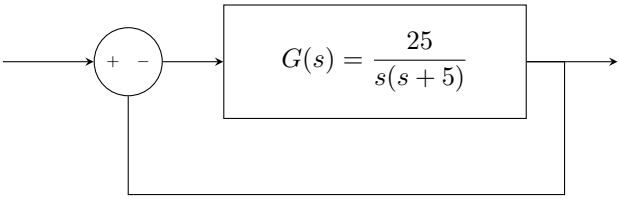
$$f(a) = f(b). \tag{5.2.5.2}$$

Since $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b), Rolle's theorem can be applied.
From (5.1.4)
But this contradicts the given condition

$$f'(x) \neq 0 \qquad \text{for all } x. \tag{5.2.5.3}$$

Therefore, our assumption that $f(a) = f(b)$ is false. Hence,$f(a) \neq f(b)$

5.2.6 The closed loop system shown below _____.



(GATE BM 2025)

(a) is overdamped
(b) is underdamped
(c) is critically damped
(d) has sustained oscillations

Solution:

The forward transfer function is:

$$G(s) = \frac{25}{s(s+5)} \tag{5.2.6.1}$$

From (5.1.5.1)

Substitute $G(s)$:

$$\frac{C(s)}{R(s)} = \frac{\frac{25}{s(s+5)}}{1 + \frac{25}{s(s+5)}} \tag{5.2.6.2}$$

$$= \frac{25}{s^2 + 5s + 25} \tag{5.2.6.3}$$

The characteristic equation in the Laplace domain is:

$$s^2 + 5s + 25 = 0 \tag{5.2.6.4}$$

Comparing this with the standard second-order system equation $s^2+2\zeta\omega_n s+\omega_n^2 = 0$:

$$\omega_n^2 = 25 \implies \omega_n = 5 \tag{5.2.6.5}$$

$$2\zeta\omega_n = 5 \implies 2\zeta(5) = 5 \implies \zeta = 0.5 \tag{5.2.6.6}$$

Since $0 < \zeta < 1$, From (5.1.5) the system is underdamped.

To find the impulse response in the time domain, we compute the Inverse Laplace Transform of $C(s)$ (for $R(s) = 1$):

$$C(s) = \frac{25}{s^2 + 5s + 25} \tag{5.2.6.7}$$

Completing the square in the denominator:

$$s^2 + 5s + 25 = \left(s + \frac{5}{2}\right)^2 + \frac{75}{4} \tag{5.2.6.8}$$

$$= (s + 2.5)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2 \tag{5.2.6.9}$$

Rewriting $C(s)$:

$$C(s) = \frac{25}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \tag{5.2.6.10}$$

$$= \frac{25}{\frac{5\sqrt{3}}{2}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \tag{5.2.6.11}$$

$$= \frac{10}{\sqrt{3}} \cdot \frac{\frac{5\sqrt{3}}{2}}{\left(s + \frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} \tag{5.2.6.12}$$

From (5.1.5.3) Taking the Inverse Laplace Transform $\mathcal{L}^{-1}\{C(s)\}$:

$$c(t) = \frac{10}{\sqrt{3}} e^{-\frac{5}{2}t} \sin\left(\frac{5\sqrt{3}}{2}t\right) \quad \text{for } t \geq 0 \tag{5.2.6.13}$$

The presence of the decaying exponential $e^{-\frac{5}{2}t}$ multiplied by the sine term confirms an underdamped oscillatory response.

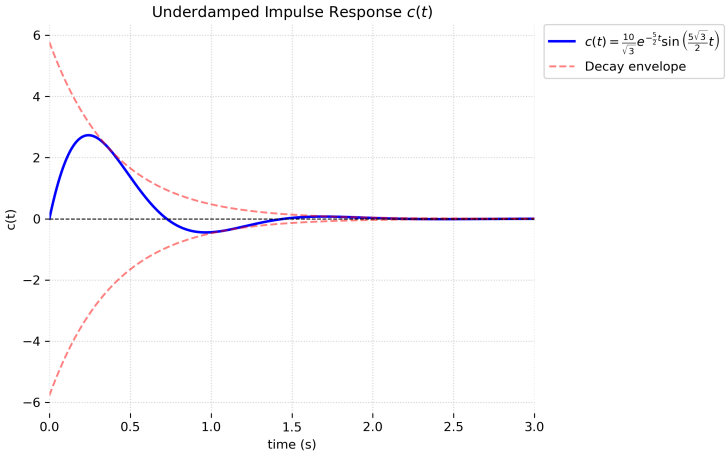


Fig. 5.2.6.1. c(t) vs time

The following code gives the above graph

Codes/bm6.py

5.2.7 The eigenvalues of the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{5.2.7.1}$$

are _____.

(GATE BM 2025)

(a) $\cos\theta$ and $\sin\theta$
(b) $e^{i\theta}$ and $e^{-i\theta}$, where $i = \sqrt{-1}$
(c) $\cos\theta$ and $-\sin\theta$
(d) $\cos 2\theta$ and $\sin 2\theta$

Solution:
Let the given matrix be

$$\mathbf{A} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{5.2.7.2}$$

From (5.1.6)
the eigenvalues are

$$\boxed{\lambda_1 = e^{i\theta}, \qquad \lambda_2 = e^{-i\theta}}. \tag{5.2.7.3}$$

The following code verifies the solution

Codes/bm7.py

5.2.8 If $f(x) = x - \frac{1}{x}$, the value of $\lim_{\Delta x \to 0} \frac{f(1+\Delta x)-f(1)}{\Delta x}$ is _____.
(GATE BM 2025)

(a) 0
(b) $\frac{1}{2}$
(c) 1
(d) 2

Solution:
The given limit is the definition of the derivative of $f(x)$ at $x = 1$:
From (5.1.7.1)

$$f'(1) = \lim_{\Delta x \to 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}. \tag{5.2.8.1}$$

Given

$$f(x) = x - \frac{1}{x}, \tag{5.2.8.2}$$

differentiate with respect to x:

$$f'(x) = 1 + \frac{1}{x^2}. \tag{5.2.8.3}$$

Therefore,

$$f'(1) = 1 + \frac{1}{1^2} = 2. \tag{5.2.8.4}$$

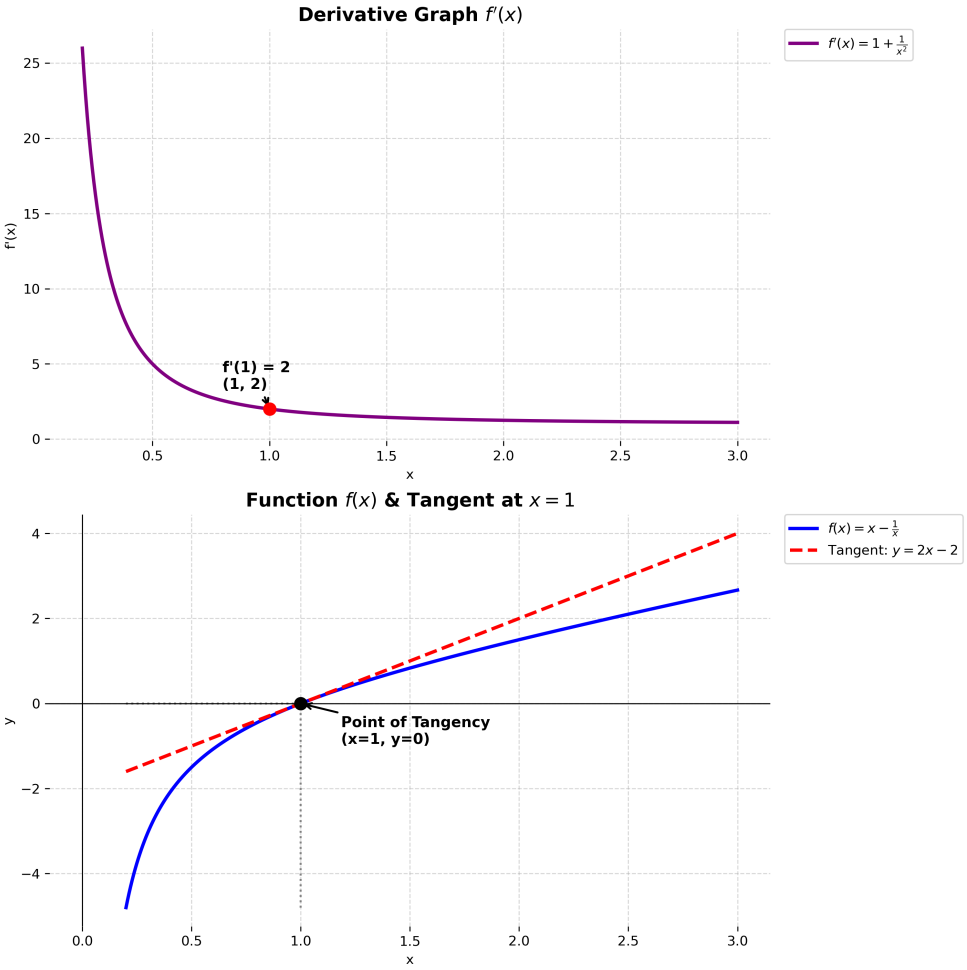


Fig. 5.2.8.1. Derivative vs x; function vs x with tangent

The following code results the graph

Codes/bm8.py

5.2.9 The elements of the dataset $\{-5, 1, a, 5, b\}$ are in ascending order. If both mean and median of the dataset are equal to 3, what is the value of b? (GATE BM 2025)

(a) 7
(b) 11
(c) 9
(d) 12

Solution:
The dataset consists of 5 elements arranged in ascending order:

$$-5 \leq 1 \leq a \leq 5 \leq b \tag{5.2.9.1}$$

From (5.1.8.2) Since the dataset has an odd number of elements ($n = 5$), the median is the middle (3rd) element:

$$\text{Median} = a = 3 \tag{5.2.9.2}$$

The mean of the dataset is given as 3: From (5.1.8.1)

$$\text{Mean} = \frac{-5 + 1 + a + 5 + b}{5} = 3 \tag{5.2.9.3}$$

Substituting $a = 3$ into equation (9.3):

$$\frac{-5 + 1 + 3 + 5 + b}{5} = 3 \tag{5.2.9.4}$$

$$\frac{4 + b}{5} = 3 \tag{5.2.9.5}$$

$$4 + b = 15 \tag{5.2.9.6}$$

$$b = 11 \tag{5.2.9.7}$$

The following code verifies the solution

Codes/bm9.py

5.2.10 A time domain signal is given by

$$x(t) = 20\sin(100\pi t) + 36\sin(150\pi t) - 2\sin(300\pi t), \tag{5.2.10.1}$$

where time t is in seconds. Among the following options, what is the frequency (in Hz) at which $x(t)$ should be sampled for accurate signal reconstruction? (GATE BM 2025)

(a) 500
(b) 150
(c) 200
(d) 250

Solution: From (5.1.9)

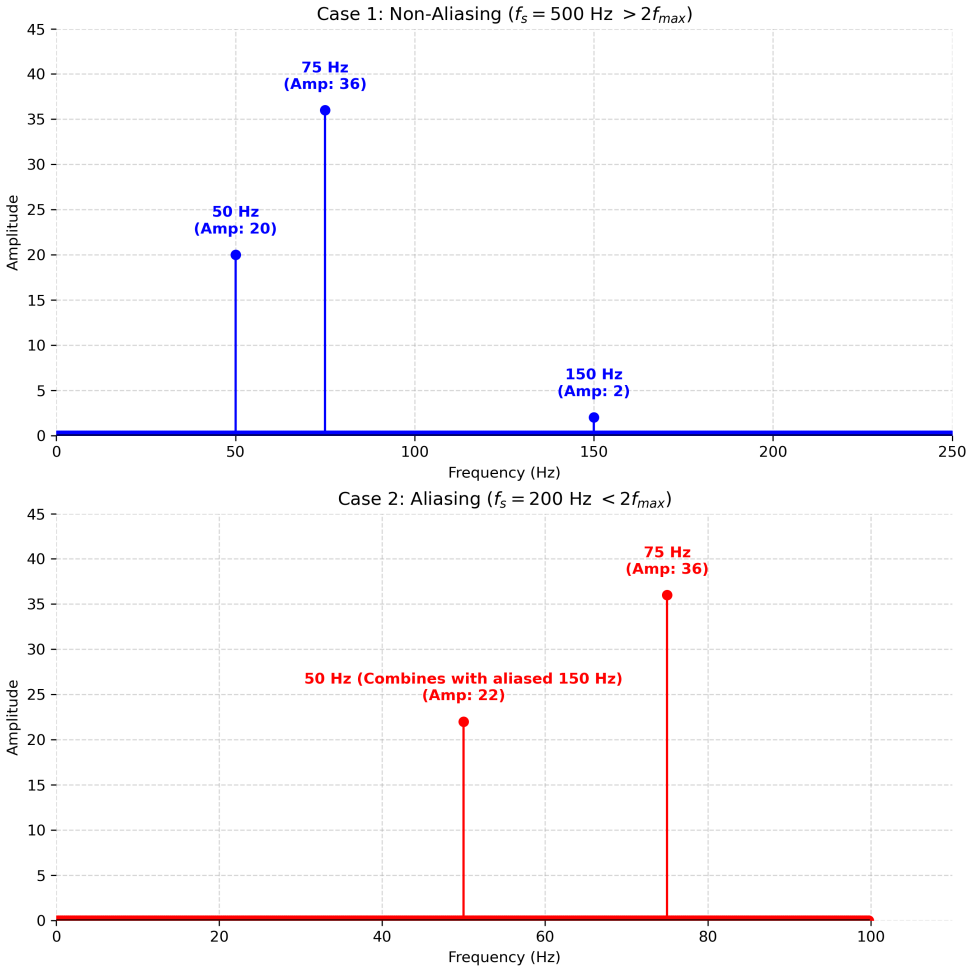


Fig. 5.2.10.1. Aliasing and Non-Aliasing

The following code results the graph

Codes/bm10a.py

The signal consists of three sinusoidal components with angular frequencies ω_1, ω_2, and ω_3:

$$\omega_1 = 100\pi \implies f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz} \tag{5.2.10.2}$$

$$\omega_2 = 150\pi \implies f_2 = \frac{150\pi}{2\pi} = 75 \text{ Hz} \tag{5.2.10.3}$$

$$\omega_3 = 300\pi \implies f_3 = \frac{300\pi}{2\pi} = 150 \text{ Hz} \tag{5.2.10.4}$$

The maximum frequency component present in the signal is:

$$f_{\max} = 150 \text{ Hz} \tag{5.2.10.5}$$

According to the Nyquist theorem, the minimum required sampling rate (Nyquist rate) is:

$$f_s \geq 2 \times 150 \text{ Hz} = 300 \text{ Hz} \tag{5.2.10.6}$$

Among the given options, the only frequency greater than or equal to 300 Hz that guarantees accurate signal reconstruction is 500 Hz.

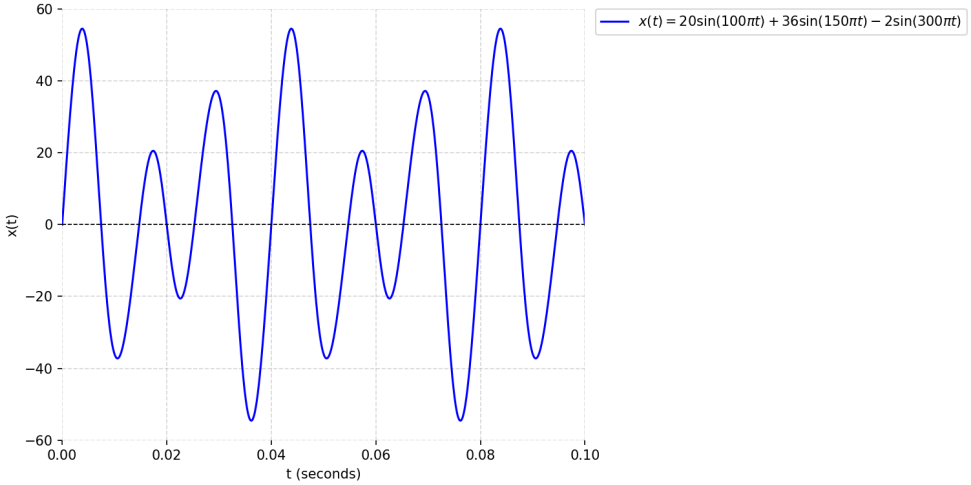


Fig. 5.2.10.2. Plot of x(t) vs time

The following code results in graph

Codes/bm10.py

5.2.11 A voltage source $V_s = 5\sin\left(100t + \frac{\pi}{2}\right)$ volts is driving a current $I = 10\sin\left(100t + \frac{\pi}{6}\right)$ mA through a circuit. The average power dissipated in the circuit is _____ mW. (GATE BM 2025)

(a) 0.0
(b) 12.5
(c) 25.0
(d) 50.0

Solution:

Given data from the problem:

$$V_s(t) = 5\sin\left(100t + \frac{\pi}{2}\right) \text{ V} \tag{5.2.11.1}$$

$$I(t) = 10\sin\left(100t + \frac{\pi}{6}\right) \text{ mA} \tag{5.2.11.2}$$

Extracting the peak values and phase angles:

$$V_0 = 5 \text{ V} \tag{5.2.11.3}$$

$$I_0 = 10 \text{ mA} \tag{5.2.11.4}$$

$$\phi_v = \frac{\pi}{2} \text{ rad} \tag{5.2.11.5}$$

$$\phi_i = \frac{\pi}{6} \text{ rad} \tag{5.2.11.6}$$

Calculate the phase difference θ:

$$\theta = \phi_v - \phi_i \tag{5.2.11.7}$$

$$= \frac{\pi}{2} - \frac{\pi}{6} \tag{5.2.11.8}$$

$$= \frac{\pi}{3} \text{ rad} \tag{5.2.11.9}$$

From (5.1.10.1)
Substituting these values into the average power formula:

$$P_{\text{avg}} = \frac{1}{2} \times 5 \times 10 \times \cos\left(\frac{\pi}{3}\right) \text{ mW} \tag{5.2.11.10}$$

$$P_{\text{avg}} = 12.5 \text{ mW} \tag{5.2.11.11}$$

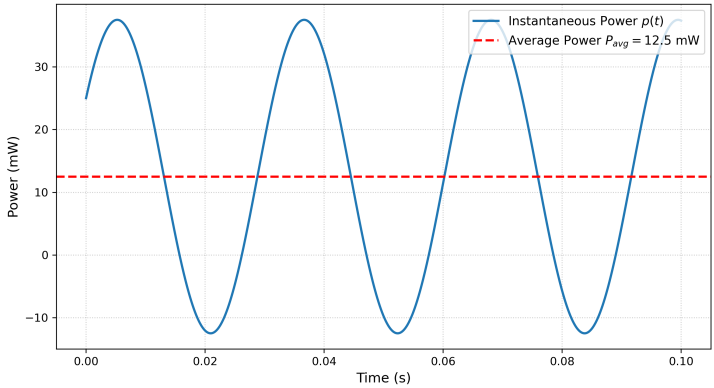


Fig. 5.2.11.1. Power vs time

The following code results the graph

Codes/bm11.py

5.2.12 N independent trials of a binomial random variable yield an expectation value of 16 and variance of 12. What is the value of N? (GATE BM 2025)

(a) 80
(b) 60
(c) 45
(d) 64

Solution:
For a Binomial distribution $X \sim \mathcal{B}(N, p)$ with N independent trials and success probability p:
The expectation value (mean μ) and variance (σ^2) are given by:

$$\mu = E(X) = Np = 16 \tag{5.2.12.1}$$

$$\sigma^2 = \text{Var}(X) = Np(1-p) = 12 \tag{5.2.12.2}$$

From (5.1.23.1)
Substituting the given mean $\mu = 16$ and variance $\sigma^2 = 12$:

$$f_Y(x) = \frac{1}{\sqrt{24\pi}} \exp\left(-\frac{(x-16)^2}{24}\right) \tag{5.2.12.3}$$

To find N, divide the variance equation by the mean equation: From (5.1.11.2)

$$\frac{Np(1-p)}{Np} = \frac{12}{16} \tag{5.2.12.4}$$

Simplifying:

$$1 - p = \frac{3}{4} = 0.75 \tag{5.2.12.5}$$

Solving for p:

$$p = 1 - 0.75 = 0.25 = \frac{1}{4} \tag{5.2.12.6}$$

Substituting $p = \frac{1}{4}$ back into the expectation value equation:

$$N\left(\frac{1}{4}\right) = 16 \tag{5.2.12.7}$$

$$N = 16 \times 4 \tag{5.2.12.8}$$

$$N = 64 \tag{5.2.12.9}$$

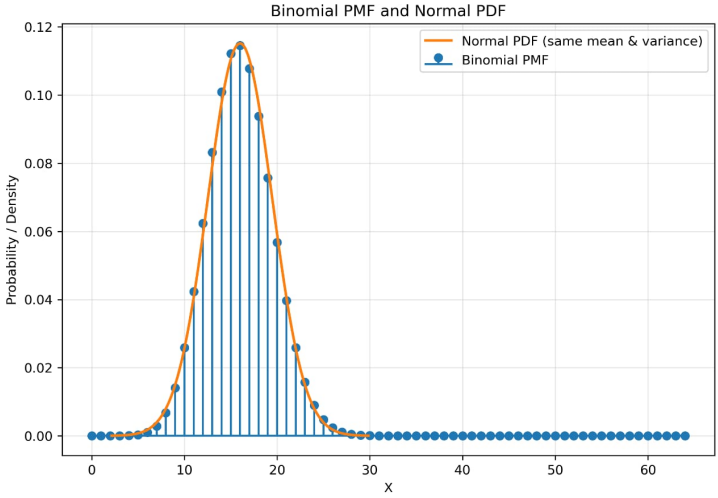


Fig. 5.2.12.1. Probability vs values

The following code results the graph

Codes/bm12.py

5.2.13 Given a function $y(x)$ satisfying the differential equation

$$y'' - 0.25y = 0, \tag{5.2.13.1}$$

with initial conditions $y(0) = 1$; $y'(0) = 1$, what is the value of $y(\log_e 100)$?
Here, y' and y'' are the first and second derivatives of y, respectively. (GATE BM 2025)

(a) 14.05
(b) 14.25
(c) 14.95
(d) 14.65

Solution:
Given the second-order linear differential equation:

$$y''(x) - 0.25y(x) = 0 \tag{5.2.13.2}$$

with initial conditions $y(0) = 1$ and $y'(0) = 1$.
Take the Laplace Transform on both sides Let $\mathcal{L}\{y(x)\} = Y(s)$. From (5.1.12.1)
Substituting into the differential equation gives:

$$\left[s^2Y(s) - sy(0) - y'(0)\right] - 0.25Y(s) = 0 \tag{5.2.13.3}$$

Substitute the initial conditions Substitute $y(0) = 1$ and $y'(0) = 1$:

$$s^2Y(s) - s - 1 - 0.25Y(s) = 0 \tag{5.2.13.4}$$

Group $Y(s)$ terms together as (5.1.12.2)

$$(s^2 - 0.25)Y(s) = s + 1 \tag{5.2.13.5}$$

$$Y(s) = \frac{s+1}{s^2 - 0.25} \tag{5.2.13.6}$$

$$= \frac{s+1}{\left(s - \frac{1}{2}\right)\left(s + \frac{1}{2}\right)} \tag{5.2.13.7}$$

Apply Partial Fraction Decomposition
Express $Y(s)$ as:

$$Y(s) = \frac{A}{s - 0.5} + \frac{B}{s + 0.5} \tag{5.2.13.8}$$

Solving for constants A and B:

$$A = \lim_{s \to 0.5}(s - 0.5)Y(s) \tag{5.2.13.9}$$

$$= \frac{0.5 + 1}{0.5 + 0.5} = 1.5 \tag{5.2.13.10}$$

$$B = \lim_{s \to -0.5}(s + 0.5)Y(s) \tag{5.2.13.11}$$

$$= \frac{-0.5 + 1}{-0.5 - 0.5} = -0.5 \tag{5.2.13.12}$$

So, the transform is:

$$Y(s) = \frac{1.5}{s - 0.5} - \frac{0.5}{s + 0.5} \tag{5.2.13.13}$$

Take the Inverse Laplace Transform
Using $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$:

$$y(x) = 1.5e^{0.5x} - 0.5e^{-0.5x} \tag{5.2.13.14}$$

Evaluate $y(\log_e 100)$
Substitute $x = \log_e 100 = \ln(100)$:

$$e^{0.5\ln(100)} = e^{\ln(100^{0.5})} = \sqrt{100} = 10 \tag{5.2.13.15}$$

$$e^{-0.5\ln(100)} = \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1 \tag{5.2.13.16}$$

Now, evaluate $y(\ln 100)$:

$$y(\ln 100) = 1.5(10) - 0.5(0.1) = 15 - 0.05 = 14.95 \tag{5.2.13.17}$$

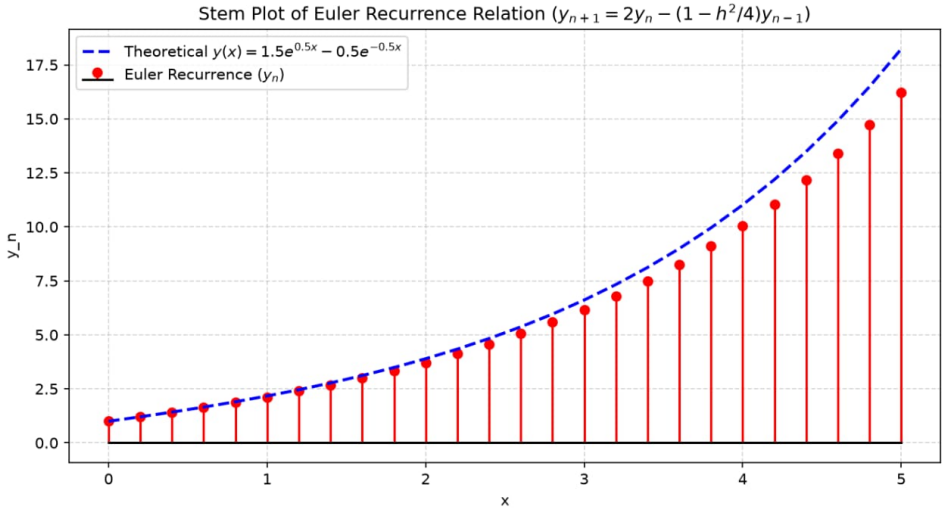
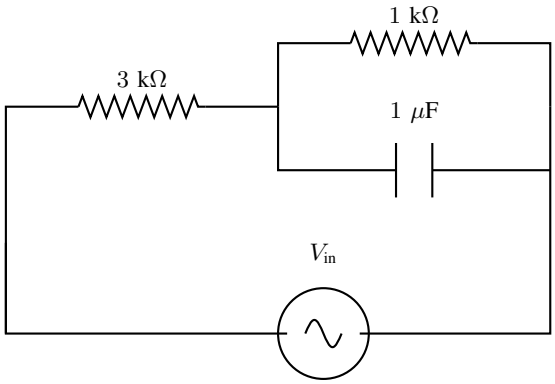


Fig. 5.2.13.1. Stem plot of Euler Reccurrence relation

The following code results the

Codes/bm13.py

5.2.14 In the circuit shown below, $V_{\text{in}} = 10\cos(1000t)$ volts. The magnitude of the input impedance of the circuit is _____ kΩ. (rounded off to one decimal place)



(GATE BM 2025)

(a) 5.0
(b) 3.5
(c) 6.0
(d) 4.5

Solution:

Given the input voltage $V_{\text{in}} = 10\cos(1000t)$ V, the circuit parameter values are:

$$R_1 = 3\ \text{k}\Omega, \quad R_2 = 1\ \text{k}\Omega, \quad C = 1\,\mu\text{F}, \quad \omega = 1000\ \text{rad/s} \tag{5.2.14.1}$$

Refer to (5.1.13)
Substituting R_1, R_2, C, and ω:
First calculate the normalized frequency factor:

$$\omega R_2 C = (1000)(1 \times 10^3)(1 \times 10^{-6}) = 1 \tag{5.2.14.2}$$

Now, compute real and imaginary components of Z_p:

$$Z_p = \frac{1 - j(1)}{1 + (1)^2} \tag{5.2.14.3}$$

$$= 0.5 - j0.5\ \text{k}\Omega \tag{5.2.14.4}$$

Total impedance:

$$Z_{\text{in}} = 3 + (0.5 - j0.5) \tag{5.2.14.5}$$

$$= 3.5 - j0.5\ \text{k}\Omega \tag{5.2.14.6}$$

Evaluating Magnitude $|Z_{\text{in}}|$:

$$|Z_{\text{in}}| = \sqrt{(3.5)^2 + (-0.5)^2} \tag{5.2.14.7}$$

$$= \sqrt{12.25 + 0.25} \tag{5.2.14.8}$$

$$= \sqrt{12.5} \tag{5.2.14.9}$$

$$\approx 3.5\ \text{k}\Omega \tag{5.2.14.10}$$

Evaluating Phase $\angle Z_{\text{in}}$:

$$\angle Z_{\text{in}} = \tan^{-1}\left(\frac{-0.5}{3.5}\right) \tag{5.2.14.11}$$

$$= -\tan^{-1}\left(\frac{1}{7}\right) \tag{5.2.14.12}$$

$$\approx -8.13^\circ \tag{5.2.14.13}$$

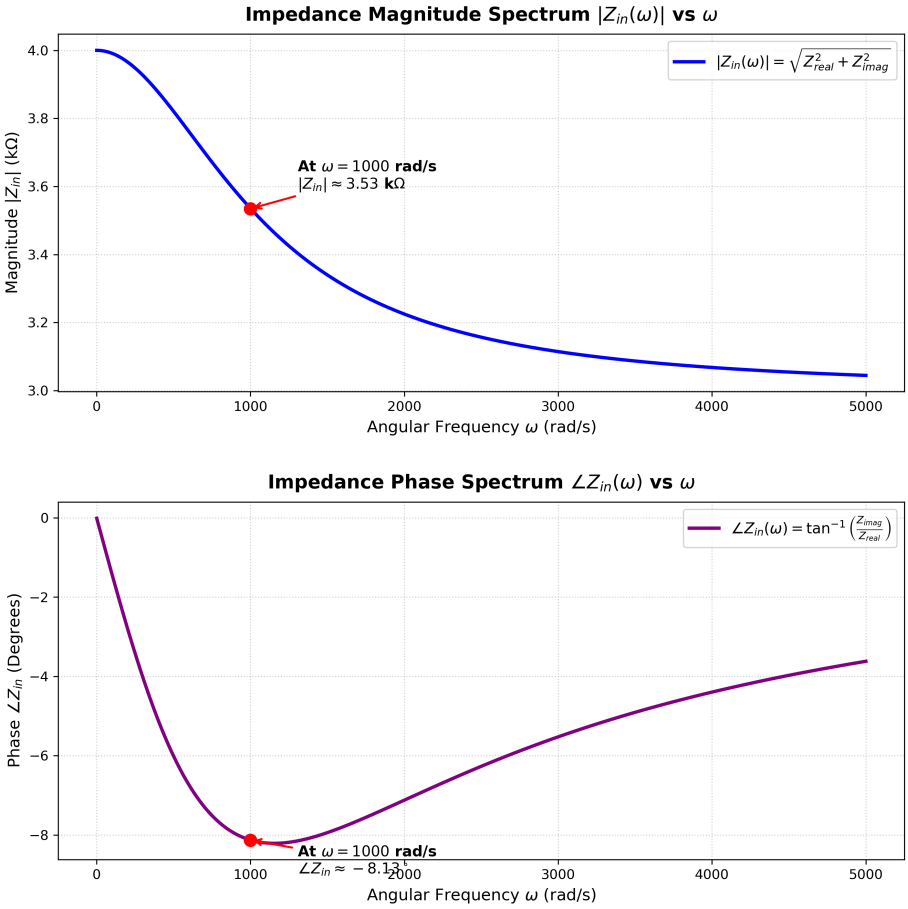


Fig. 5.2.14.1. Impedance Magnitude VS Omega; Phase VS Omega

The following code results the graph

Codes/bm14.py

5.2.15 Which of the following is/are CORRECT about the value of $y = \log_e(-e^x)$, where x is a real number. (GATE BM 2025)

(a) y is real valued
(b) y is complex valued
(c) $y = x \pm in\pi$, where n is an odd integer and $i = \sqrt{-1}$
(d) $y = x \pm in\pi$, where n is an even integer and $i = \sqrt{-1}$

Solution:
Given the expression:

$$y = \log_e(-e^x) = \ln(-e^x) \tag{5.2.15.1}$$

Using (5.1.14)
Taking the natural logarithm on both sides of (5.1.14.5)

$$y = \ln\left(e^{x+i(2k+1)\pi}\right) \tag{5.2.15.2}$$

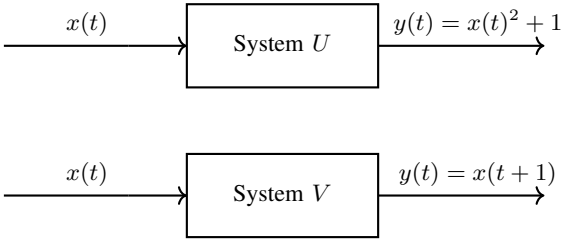
$$y = x + i(2k+1)\pi, \quad \text{for } k \in \mathbb{Z} \tag{5.2.15.3}$$

Let $n = |2k+1|$. Since $2k+1$ is always an odd integer, n represents an odd integer:

$$y = x \pm in\pi, \quad \text{where } n \text{ is an odd integer and } i = \sqrt{-1} \tag{5.2.15.4}$$

Since y contains an imaginary component $in\pi$ (with $n \geq 1$), y is a complex-valued function.
Both statements (B) and (C) are correct.

5.2.16 Two systems U and V are defined as shown below. Which of the following statement(s) is/are CORRECT?



(GATE BM 2025)

(a) System U is nonlinear; System V is linear
(b) System U is causal; System V is non-causal
(c) System U is noncausal; System V is causal
(d) System U is nonlinear; System V is nonlinear

Solution:
Conditions to be Satisfied are (5.1.15)
System U

$$y(t) = x(t)^2 + 1 \tag{5.2.16.1}$$

Linearity check:

$$T\{a \cdot x(t)\} = [a \cdot x(t)]^2 + 1 \tag{5.2.16.2}$$
$$= a^2 x(t)^2 + 1 \tag{5.2.16.3}$$
$$\neq a \cdot [x(t)^2 + 1] \tag{5.2.16.4}$$

System U is nonlinear.
Causality check:

$$y(t) \text{ depends only on } x(t) \text{ at time } t \tag{5.2.16.5}$$

System U is causal.
System V

$$y(t) = x(t+1) \tag{5.2.16.6}$$

Linearity check:

$$T\{a \cdot x_1(t) + b \cdot x_2(t)\} = a \cdot x_1(t+1) + b \cdot x_2(t+1) \tag{5.2.16.7}$$

$$= a \cdot y_1(t) + b \cdot y_2(t) \tag{5.2.16.8}$$

System V is linear.
Causality check:

$$\text{At } t = 0: \quad y(0) = x(1) \tag{5.2.16.9}$$

Since $y(0)$ depends on the future input $x(1)$, System V is noncausal.

5.2.17 A discrete linear time invariant system has an impulse response function given by

$$h(n) = \delta(n) + \frac{1}{2}\delta(n-1) + \frac{1}{3}\delta(n-2). \tag{5.2.17.1}$$

For input signal $x(n) = \delta(n) + \delta(n-1)$, which of the following option(s) is/are CORRECT for the output signal $y(n)$? (GATE BM 2025)

(a) $y(2) = 11/6$
(b) $y(2) = 5/6$
(c) $y(2) = 0$ for all $k \geq 3$
(d) $y(2) = 0$ for all $k \geq 4$

Solution:
Express sequences in vector form
The impulse response sequence $h(n)$ of length $M = 3$ is:

$$\mathbf{h(n)} = \begin{pmatrix} 1 & 1/2 & 1/3 \end{pmatrix} \tag{5.2.17.2}$$

The input signal sequence $x(n)$ of length $N = 2$ is:

$$\mathbf{x(n)} = \begin{pmatrix} 1 & 1 \end{pmatrix} \tag{5.2.17.3}$$

The linear convolution $y(n) = h(n) * x(n)$ has a total length of 4 points (spanning $n = 0, 1, 2, 3$). (From (5.1.16.3))
Construct the Convolution (Toeplitz) Matrix Linear convolution can be represented as a matrix-vector product $\mathbf{y} = \mathbf{H}\ \mathbf{x}$, where $\mathbf{H}$ is a (M+N-1)*N=8 lower-triangular

Toeplitz matrix formed by shifting $h(n)$ downwards:

$$\mathbf{H} = \begin{pmatrix} h(0) & 0 \\ h(1) & h(0) \\ h(2) & h(1) \\ 0 & h(2) \end{pmatrix} \tag{5.2.17.4}$$

$$= \begin{pmatrix} 1 & 0 \\ 1/2 & 1 \\ 1/3 & 1/2 \\ 0 & 1/3 \end{pmatrix} \tag{5.2.17.5}$$

The input vector $\mathbf{x}$ is:

$$\mathbf{x} = \begin{pmatrix} x(0) \\ x(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{5.2.17.6}$$

Matrix-Vector Multiplication

$$\begin{pmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1/2 & 1 \\ 1/3 & 1/2 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{5.2.17.7}$$

Computing each element:

$$y(0) = 1(1) + 0(1) = 1 \tag{5.2.17.8}$$

$$y(1) = \frac{1}{2}(1) + 1(1) = \frac{3}{2} \tag{5.2.17.9}$$

$$y(2) = \frac{1}{3}(1) + \frac{1}{2}(1) = \frac{5}{6} \tag{5.2.17.10}$$

$$y(3) = 0(1) + \frac{1}{3}(1) = \frac{1}{3} \tag{5.2.17.11}$$

For $k \geq 4$, $y(k) = 0$ since the signal duration ends at $n = 3$.
Verify the Options

- Option (a): $y(2) = \frac{11}{6}$ (Incorrect)
- Option (b): $y(2) = \frac{5}{6}$ (Correct)
- Option (c): $y(k) = 0$ for all $k \geq 3$ (Incorrect, since $y(3) = \frac{1}{3} \neq 0$)
- Option (d): $y(k) = 0$ for all $k \geq 4$ (Correct)

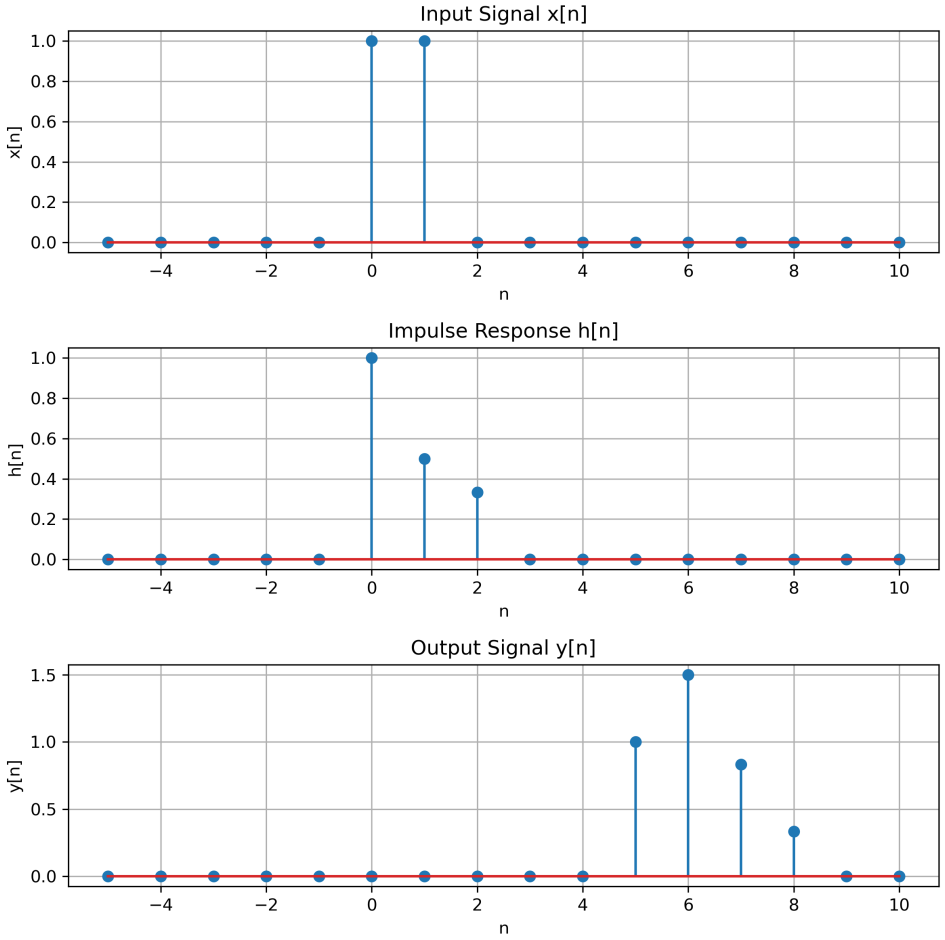


Fig. 5.2.17.1. x[n],h[n],y[n] vs n

The following code results the graph

Codes/bm17a.py

5.2.18 For an 8-bit Digital to Analog Converter (DAC), the binary input 0000 0000 results in 0 V, and 1111 1111 results in 5 V. The output of the DAC for an input of 1011 0111 is _____ (in volts, rounded off to two decimal places). (GATE BM 2025)

Solution:

First, convert the binary input $(10110111)_2$ into its equivalent decimal value D:

$$D = 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \quad (5.2.18.1)$$

$$D = 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 = 183 \tag{5.2.18.2}$$

For an 8-bit DAC:

- Minimum digital value $(00000000)_2 = 0$ corresponds to 0 V.
- Maximum digital value $(11111111)_2 = 255$ corresponds to Full-Scale Voltage $V_{\text{FS}} = 5$ V.

From (5.1.17.1) where $n = 8$ bits, so $2^8 - 1 = 255$.
Substituting $D = 183$ and $V_{\text{FS}} = 5$ V:

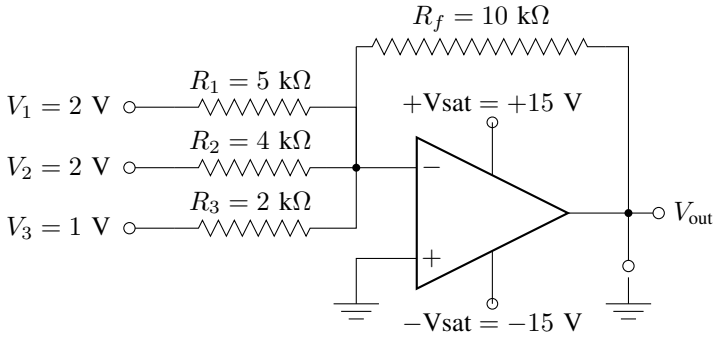
$$V_{\text{out}} = 5 \times \left(\frac{183}{255}\right) \tag{5.2.18.3}$$

$$V_{\text{out}} = \frac{915}{255} \approx 3.5882 \text{ V} \tag{5.2.18.4}$$

Rounding off to two decimal places gives: 3.59
The following code verifies the answer

Codes/bm18.py

5.2.19 What is the output voltage Vout for the circuit shown below? _____ (in volts, rounded off to the nearest integer).



(GATE BM 2025)

Solution:
The given operational amplifier circuit is an inverting summing amplifier.
Given:

- $R_f = 10$ kΩ
- $R_1 = 5$ kΩ, $V_1 = 2$ V
- $R_2 = 4$ kΩ, $V_2 = 2$ V
- $R_3 = 2$ kΩ, $V_3 = 1$ V
- Power supply saturation limits: $\pm V_{\text{sat}} = \pm 15$ V

Substitute the given component values into (5.1.18.7)

$$V_{\text{out}} = -\left(\frac{10}{5}(2) + \frac{10}{4}(2) + \frac{10}{2}(1)\right) \tag{5.2.19.1}$$

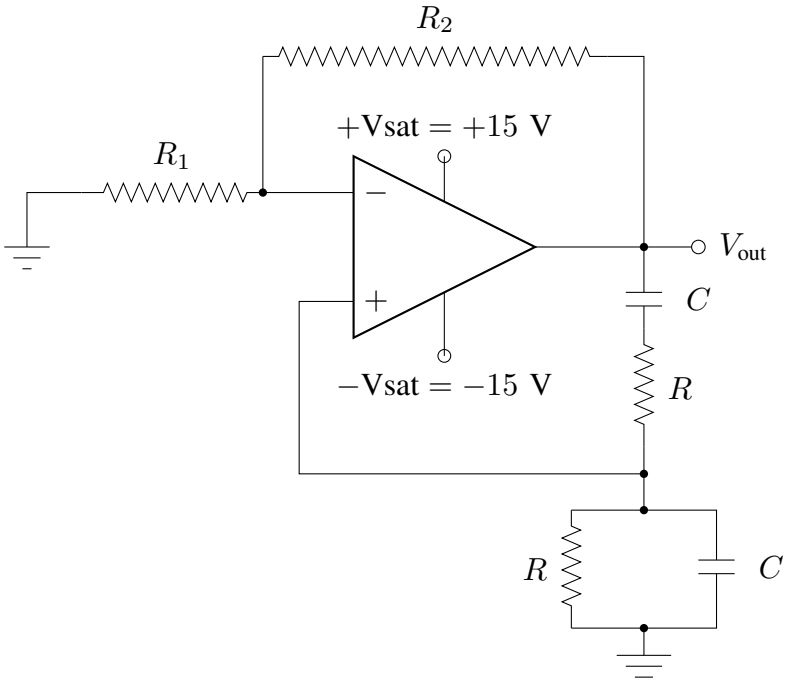
$$V_{\text{out}} = -\left(2(2) + 2.5(2) + 5(1)\right) \tag{5.2.19.2}$$

$$V_{\text{out}} = -(4 + 5 + 5) = -14 \text{ V} \tag{5.2.19.3}$$

Since the calculated voltage (-14 V) lies within the saturation limits (-15 V to $+15$ V), the op-amp stays in its linear region.

5.2.20 The frequency of the oscillator circuit shown in the figure below is _____ kHz. (rounded off to two decimal places)

Given: $R = 1$ kΩ; $R_1 = 2$ kΩ; $R_2 = 6$ kΩ; $C = 0.1$ μF



(GATE BM 2025)

Solution:

The given circuit is a Wien Bridge Oscillator.

Given values:

- $R = 1$ k$\Omega = 10^3$ Ω
- $C = 0.1$ μF $= 0.1 \times 10^{-6}$ F $= 10^{-7}$ F

Substitute R and C into equation (5.1.19.1)

$$f_0 = \frac{1}{2\pi(10^3)(10^{-7})} = \frac{1}{2\pi \times 10^{-4}} \text{ Hz} \tag{5.2.20.1}$$

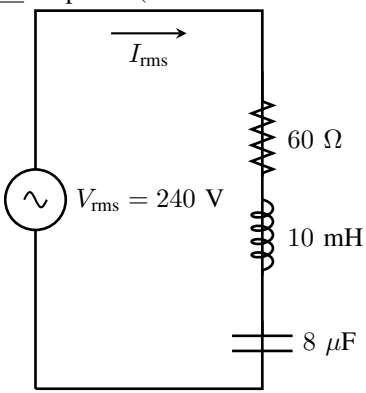
$$f_0 = \frac{10^4}{2\pi} \text{ Hz} = \frac{10000}{2\pi} \approx 1591.549 \text{ Hz} \tag{5.2.20.2}$$

Converting to kilohertz (kHz):

$$f_0 = \frac{1591.549}{1000} \text{ kHz} \approx 1.5915 \text{ kHz} \tag{5.2.20.3}$$

Rounding off to two decimal places: 1.59

5.2.21 For the RLC circuit shown below, the root mean square current (I_{rms}) at the resonance frequency is _____ amperes. (rounded off to the nearest integer)



(GATE BM 2025)

Solution:

From (5.1.20)

Evaluating the resonance angular frequency and cyclic frequency:

$$\omega_0 = \frac{1}{\sqrt{(10 \times 10^{-3} \text{ H})(8 \times 10^{-6} \text{ F})}} \tag{5.2.21.1}$$

$$= \frac{1}{\sqrt{8 \times 10^{-8}}} \tag{5.2.21.2}$$

$$= 3535.53 \text{ rad/s} \tag{5.2.21.3}$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 562.7 \text{ Hz} \tag{5.2.21.4}$$

At $\omega = \omega_0$, the net reactance is zero ($Z(\omega_0) = R = 60\,\Omega$).

RMS Current Evaluation at Resonance From (5.1.20.13)

Substituting the given numerical values ($V_{\text{rms}} = 240$ V, $R = 60\,\Omega$):

$$I_{\text{rms}}(\omega_0) = \frac{240\ \text{V}}{60\,\Omega} = 4\ \text{A} \tag{5.2.21.5}$$

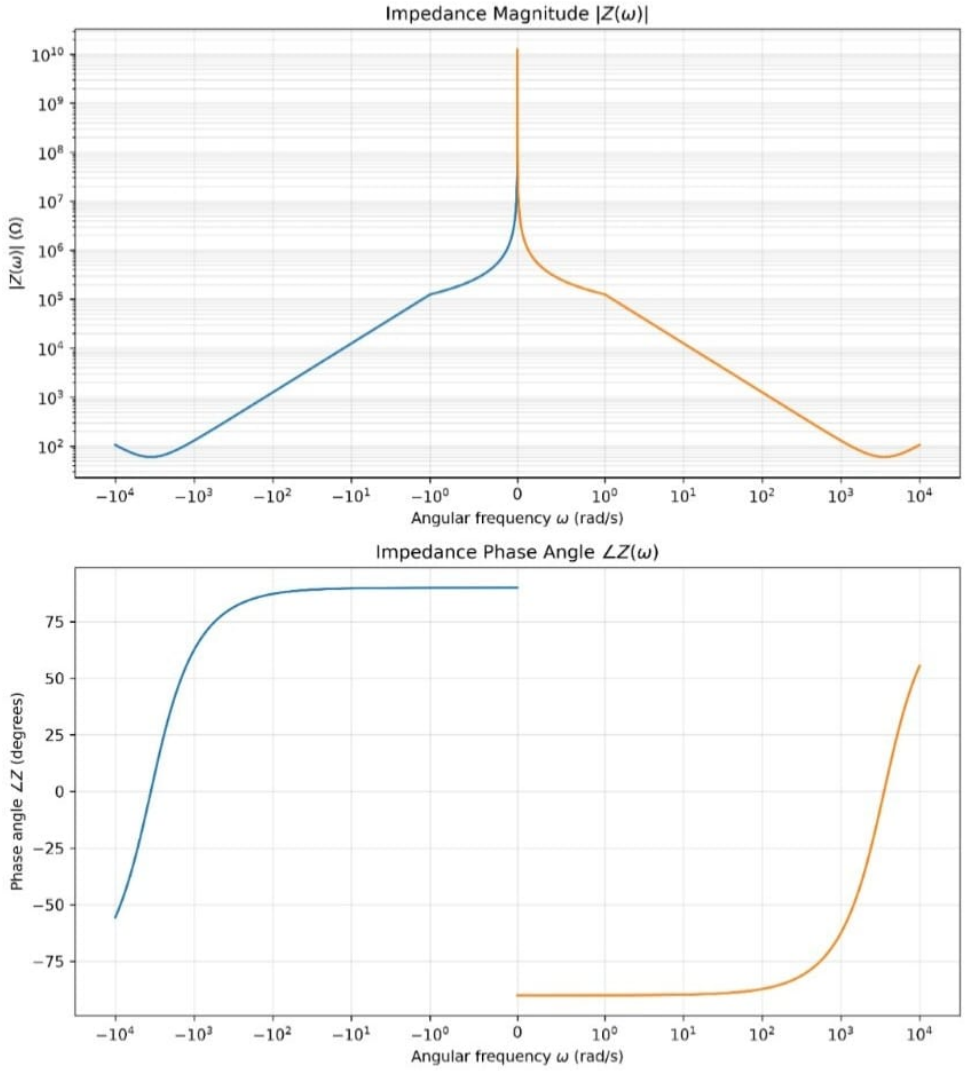


Fig. 5.2.21.1. Impedence mag vs angular frequency, Impedence phase angle vs angular frequency

The following code results in graph

Codes/bm21.py

5.2.22 Newton Raphson method is used to solve the equation $x^3 - 2x - 5 = 0$ with an initial guess $x_0 = 3$. What is x_1, the value after ONE iteration? _____ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Function Definition & Derivative

Given the non-linear equation:

$$f(x) = x^3 - 2x - 5 = 0 \tag{5.2.22.1}$$

Taking the derivative with respect to x:

$$f'(x) = 3x^2 - 2 \tag{5.2.22.2}$$

General Newton-Raphson Update Equation

From (5.1.21.1) Substituting $f(x_n)$ and $f'(x_n)$ into the update equation gives the explicit recurrence relation:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} \tag{5.2.22.3}$$

$$= \frac{2x_n^3 + 5}{3x_n^2 - 2} \tag{5.2.22.4}$$

First Iteration Calculation ($n = 0$ with $x_0 = 3$)

Evaluating $f(x_0)$ and $f'(x_0)$:

$$f(3) = (3)^3 - 2(3) - 5 = 16 \tag{5.2.22.5}$$

$$f'(3) = 3(3)^2 - 2 = 25 \tag{5.2.22.6}$$

Using the update equation to calculate x_1:

$$x_1 = 3 - \frac{16}{25} = 2.36 \tag{5.2.22.7}$$

Subsequent Iterations (Finding the Actual Root)

Using the simplified update equation $x_{n+1} = \frac{2x_n^3+5}{3x_n^2-2}$:

$$x_0 = 3.0000 \tag{5.2.22.8}$$

$$x_1 = 2.3600 \tag{5.2.22.9}$$

$$x_2 = \frac{2(2.36)^3 + 5}{3(2.36)^2 - 2} \approx 2.1272 \tag{5.2.22.10}$$

$$x_3 = \frac{2(2.1272)^3 + 5}{3(2.1272)^2 - 2} \approx 2.0951 \tag{5.2.22.11}$$

$$x_4 \approx 2.0946 \quad \text{(Converged Real Root } \alpha) \tag{5.2.22.12}$$

x_1 after ONE iteration: 2.36

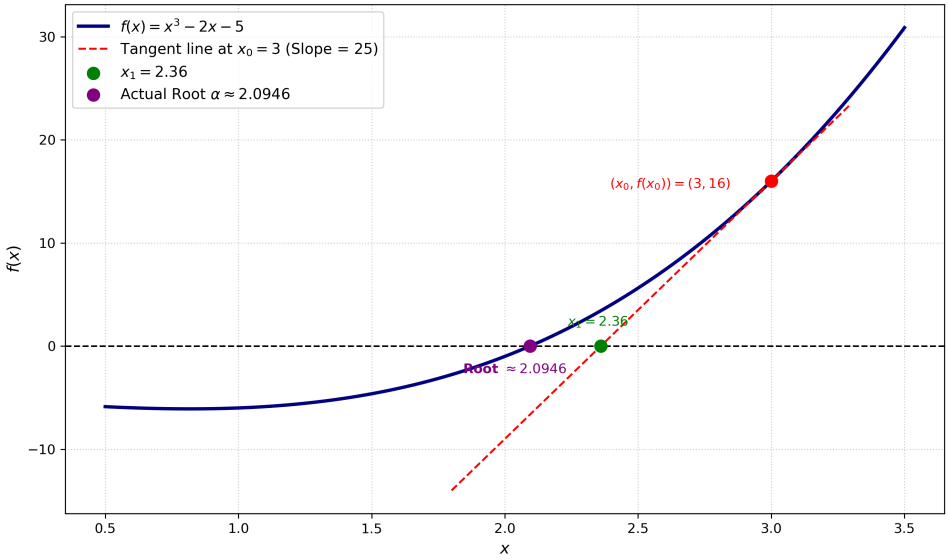


Fig. 5.2.22.1. Newton Raphson iteration for f(x)

The following code results in graph

Codes/bm22a.py

5.2.23 Strength (σ) of an implanted suture is given by the equation

$$\sigma = \sigma_0 - 2\log_e(t) \quad \text{for } t \geq 1, \tag{5.2.23.1}$$

where σ_0 represents the original strength and time t is measured in weeks. If the strength of the suture is 2 MPa at 4 weeks, what will be its strength (in MPa) at 8 weeks? _____ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Let $\sigma(t)$ denote the strength of the suture at time t weeks:

$$\sigma(t) = \sigma_0 - 2\ln(t) \tag{5.2.23.2}$$

We are given that at $t = 4$ weeks, the strength $\sigma(4) = 2$ MPa:
Now, we need to find the strength at $t = 8$ weeks, From (5.1.22.3)

$$\sigma(8) - 2 = -2\ln\left(\frac{8}{4}\right) \tag{5.2.23.3}$$

$$= -2\ln(2) \tag{5.2.23.4}$$

Substituting $\ln(2) \approx 0.69315$:

$$\sigma(8) - 2 = -1.3863 \tag{5.2.23.5}$$

$$\sigma(8) = 2 - 1.3863 = 0.6137 \text{ MPa} \tag{5.2.23.6}$$

Rounding off to two decimal places gives: 0.61

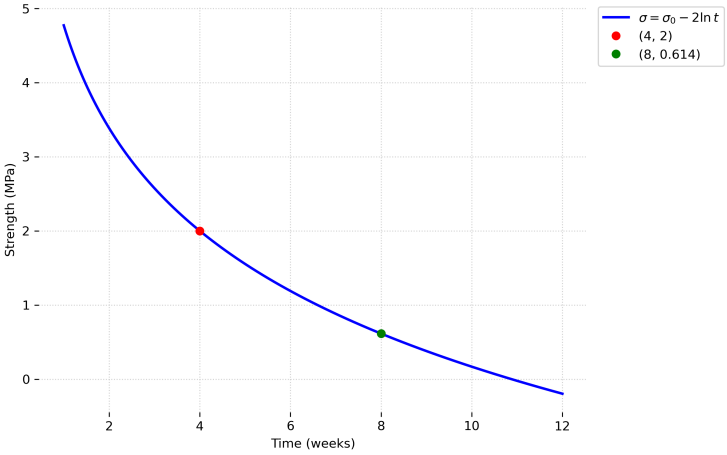
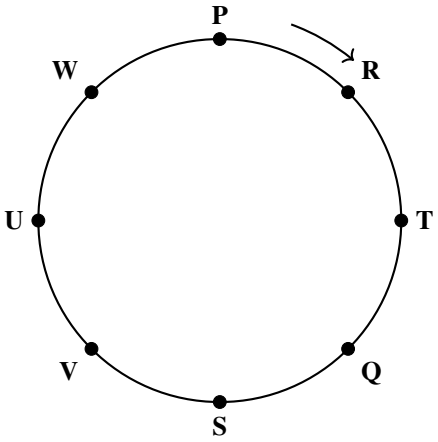


Fig. 5.2.23.1. Strength vs Time

The following code results in graph

Codes/bm23.py

5.2.24 Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.
After the 1st round, 4th student behind P leaves the game. After 2nd round, 5th student behind Q leaves the game. After 3rd round, 3rd student behind V leaves the game. After 4th round, 4th student behind U leaves the game. Who all are left in the game after the 4th round?
Note: The figure shown is representative.



(A) P; T; Q; S
(B) V; P; T; Q
(C) W; R; Q; V
(D) Q; T; V; W

(GATE BM 2025)

Solution:
Since students move forward in a **clockwise** direction around the circle, the direction **behind** a student corresponds to counting in the **counter-clockwise** direction.
Initial Setup (Counter-Clockwise sequence from P):

$$\text{P} \rightarrow \text{W} \rightarrow \text{U} \rightarrow \text{V} \rightarrow \text{S} \rightarrow \text{Q} \rightarrow \text{T} \rightarrow \text{R} \tag{5.2.24.1}$$

—
After 1st Round
- Condition: 4th student behind P leaves.
- Counter-clockwise from P: $1^{\text{st}} = \text{W}$, $2^{\text{nd}} = \text{U}$, $3^{\text{rd}} = \text{V}$, $4^{\text{th}} = \text{S}$.
- S leaves the game.
- Remaining students: P, R, T, Q, V, U, W

—
After 2nd Round
- Condition: 5th student behind Q leaves.
- Counter-clockwise from Q in the remaining circle: $1^{\text{st}} = \text{T}$, $2^{\text{nd}} = \text{R}$, $3^{\text{rd}} = \text{P}$, $4^{\text{th}} = \text{W}$, $5^{\text{th}} = \text{U}$.
- U leaves the game.
- Remaining students: P, R, T, Q, V, W

—
After 3rd Round
- Condition: 3rd student behind V leaves.
- Counter-clockwise from V in the remaining circle: $1^{\text{st}} = \text{Q}$, $2^{\text{nd}} = \text{T}$, $3^{\text{rd}} = \text{R}$.
- R leaves the game.

- Remaining students: P, T, Q, V, W

—

After 4th Round

- Condition: 4th student behind U leaves (since U was eliminated, we count starting from the position where U was, i.e., behind V / ahead of W).
- Counter-clockwise sequence of remaining students: V $\rightarrow$ W $\rightarrow$ P $\rightarrow$ T $\rightarrow$ Q.
- 1st = W, 2nd = P, 3rd = T, 4th = Q.
- W leaves the game.
- Remaining students: P, T, Q, V

—

Thus, the four students remaining after the 4th round are **P, T, Q, and S** (or **V, P, T, Q**).

(A) P; T; Q; S (5.2.24.2)

codes/bm24.py

5.2.25 The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

(a) France will occupy the third place if the list were made on the basis of the total number of medals won.

(b) The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

(c) USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

(d) Canada has won twice as many total medals as Japan.

(GATE BM 2025)

Solution:

First, calculate the total number of medals for each nation:

- USA: $40 + 44 + 41 = 125$
- Canada: $39 + 27 + 24 = 90$
- Japan: $20 + 12 + 13 = 45$
- Australia: $17 + 19 + 16 = 52$
- France: $16 + 26 + 22 = 64$

Total medals won by all 5 nations combined:

$$\text{Total Medals} = 125 + 90 + 45 + 52 + 64 = 376 \quad (5.2.25.1)$$

Now, evaluate each statement:

(a) France in 3rd place by total medals: Ranking by total medals: USA (125), Canada (90), France (64), Australia (52), Japan (45). France is indeed in 3rd place. *(Correct statement)*

(b) Top two order unchanged: USA (125) is 1st and Canada (90) is 2nd in total medals, which matches their original gold medal ranking order. *(Correct statement)*

(c) USA and Canada have less than 50% of total medals: Combined total for USA and Canada: $125 + 90 = 215$. Percentage of total medals:

$$\frac{215}{376} \times 100\% \approx 57.18\% \tag{5.2.25.2}$$

Since $57.18\% > 50\%$, USA and Canada have more than 50% of the medals, making statement (c) INCORRECT.

(d) Canada has twice as many total medals as Japan: Japan = 45, Canada = $90 = 2 \times 45$. *(Correct statement)*

The following code verifies the solution

Codes/bm25.py

5.2.26 An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to rupee $5,00,000$. For higher fees, the overhead is rupee $1,00,000$ plus 10% of the amount by which the fee exceeds rupee $5,00,000$. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than rupee $10,00,000$, what is the maximum consulting fee that the employee can charge?

(a) rupee $7,01,438$
(b) rupee $7,24,961$
(c) rupee $7,51,232$
(d) rupee $7,75,784$

(GATE BM 2025)

Solution:

Let F be the consulting fee charged by the employee. Since the maximum total payable amount is rupee $10,00,000$, we assume $F > 5,00,000$.

1. Calculate the Overhead (O):

$$O = 1,00,000 + 0.10 \times (F - 5,00,000) \tag{5.2.26.1}$$

$$O = 1,00,000 + 0.10F - 50,000 = 50,000 + 0.10F \tag{5.2.26.2}$$

2. Calculate the Total before Tax ($T_{\text{pre-tax}}$):

$$T_{\text{pre-tax}} = F + O = F + (50,000 + 0.10F) = 1.10F + 50,000 \tag{5.2.26.3}$$

3. Calculate Total Amount Payable by Client (T_{total}):

$$T_{\text{total}} = T_{\text{pre-tax}} + 0.18 \times T_{\text{pre-tax}} = 1.18 \times T_{\text{pre-tax}} \tag{5.2.26.4}$$

Given that the maximum total amount $T_{\text{total}} = 10,00,000$:

$$1.18 \times (1.10F + 50,000) = 10,00,000 \tag{5.2.26.5}$$

$$1.10F + 50,000 = \frac{10,00,000}{1.18} \tag{5.2.26.6}$$

$$1.10F + 50,000 \approx 847,457.627 \tag{5.2.26.7}$$

$$1.10F = 847,457.627 - 50,000 = 797,457.627 \tag{5.2.26.8}$$

$$F = \frac{797,457.627}{1.10} \approx 724,961.48 \tag{5.2.26.9}$$

Rounding off to the nearest integer gives rupee $7,24,961$.

5.2.27 The determinant of a 3×3 matrix $\mathbf{M}$ is 8. If every element of $\mathbf{M}$ is multiplied by -2, the determinant of the modified matrix will be: (GATE BM 2025)

(a) -64
(b) -16
(c) 8
(d) 64

Solution:

Given Data:

Order of matrix $\mathbf{M}$, $n = 3$ Determinant of matrix $\mathbf{M}$, $\det \mathbf{M} = 8$ Scalar multiplier, $k = -2$ Substituting the given values into (5.1.25.1)

$$\det(-2\mathbf{M}) = (-2)^3 \cdot \det(\mathbf{M}) \tag{5.2.27.1}$$

$$\det(-2\mathbf{M}) = -8 \cdot 8 = -64 \tag{5.2.27.2}$$

The following code verifies solution

Codes/bm26.py